॥ श्री गणेश प्रसन्न ॥

Crazy 4

Papers 1–4 combined · Sections 14–17

All four answer books merged section by section; overlapping answers are printed once with every fact kept, tagged P1–P4 by where they were asked.

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Section 14 · Adrenal & Retroperitoneum

Long answer questions (LAQ)

LAQ 1 Phaeochromocytoma: adrenal anatomy, hormones and perioperative management

P1 · P2 **merged from 2**

Asked as: P1 LAQ "Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma" (Practice Set 6 Paper 1 Q1) · P2 SAQ "Phaeochromocytoma: overview, pre-operative and intra-operative management" (asked as "Pre operative management of pheochromocytoma", Summer 2020 Paper II Q3c; "Intra op mgt of pheo", Practice Set 7 Paper II Q3c; "Pheochromocytoma", Practice Set 1 Paper II Q3d)

Assumed meaning: "arenal" = adrenal glands.

Adrenal anatomy

Gross

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

Feature	Right	Left
Shape, level	Triangular; higher	Crescentic; lower (to L1)
Relations	Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind	Pancreas body and splenic vessels anteriorly, aorta medially, crus behind
Vein	**Short (<1 cm), into posterior IVC**: tears easily	**Longer, into left renal vein**, often joined by inferior phrenic vein

Blood supply

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta) and inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

Nerves: preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in the coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

Embryology: cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")

Zone	Share	Product	Main control
Glomerulosa	~15% of cortex	Aldosterone	Angiotensin II, potassium
Fasciculata	~80%	Cortisol (up to 20 mg/day)	ACTH (adrenocorticotropic hormone)
Reticularis	5–7%	DHEA (dehydroepiandrosterone), DHEA-S, androstenedione	ACTH
Medulla	~10% of gland	Adrenaline 80%, noradrenaline 19%, dopamine 1%	Sympathetic nerves

Steps in adrenal hormone production

1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product: `Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only) Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH-> CORTISOL (fasciculata) 17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)`
- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) drives ACTH, which drives cortisol, with negative feedback and a morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium control aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

2. Catecholamines (medulla) `Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine -dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline`

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and the liver form VMA (vanillylmandelic acid).

Phaeochromocytoma: definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy.
- **VHL (von Hippel–Lindau)**: normetanephrine only.
- **NF1 (neurofibromatosis type 1)**.
- **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**. (SDHB/SDHD = succinate dehydrogenase subunits.)
- Offer genetic testing to all: about one-third hereditary (MEN2, VHL, NF1, SDHB/SDHD).

Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

Why preparation is needed

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied.
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

Pre-operative step 1: confirm, localise, test genes

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.
- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging).
- **Functional imaging** if extra-adrenal or hereditary: ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); FDG (fluorodeoxyglucose) PET (best for SDHB); ^{123}I-MIBG (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.
- Genetic testing (about one-third hereditary).

Pre-operative step 2: cardiac assessment

- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy; joint plan with the anaesthetist.

Pre-operative step 3: drug blockade (7–14 days)

Drug	Action	Points
Phenoxybenzamine (first line)	Irreversible non-selective α-blocker	10 mg twice daily, increased every 2–3 days (titrated); about 1 mg/kg total
Doxazosin, prazosin	Selective α1-blockers	Fewer side effects

Drug	Action	Points
Calcium channel blocker	Smooth muscle relaxation	Alternative strategy or add-on
β1-blocker (atenolol, metoprolol)	Controls tachycardia	**Only after adequate α-blockade**
Metyrosine	Blocks tyrosine hydroxylase	Add-on; sedation, extrapyramidal effects

- **α-blockade for all**, 7–14 days.
- **Never β before α:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed, which causes crisis.
- **Target:** seated BP (blood pressure) 120–130/80 mmHg; a mild postural drop (systolic >90 mmHg) shows adequate blockade.

Pre-operative step 4: volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV (intravenous) fluid the night before. This prevents post-excision hypotension.

Pre-operative step 5: day of surgery

- Last phenoxybenzamine dose the night before surgery; omit the morning dose.

Intra-operative management

Prerequisites

- α-blockade 7–14 days (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG and echocardiography; joint plan with the anaesthetist.

Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate the intubation response.
- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

Haemodynamic control

Event	Action
Hypertensive surge	**Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine
Tachyarrhythmia	**Esmolol**, lidocaine; amiodarone if needed
Hypotension after vein ligation	**Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory
Hypoglycaemia (insulin rebound)	Dextrose infusion, glucose checks

Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2, VHL and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

Post-operative management

- HDU (high dependency unit) overnight: blood pressure, heart rate, glucose.
- Fluids ± vasopressors for **hypotension**.
- Glucose monitoring for **rebound hypoglycaemia** (insulin no longer suppressed).
- Steroid cover after bilateral adrenalectomy.

Follow-up and metastatic disease

- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).
- Malignancy is defined only by **metastases**. Treat with metastasectomy, ^{131}I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).

Diagrams to draw

- Posterior view of both adrenals with the three arteries and the two veins.
- Flowchart: metanephrines, imaging, α-blockade, β-blockade, salt and fluids, surgery.
- Intra-operative BP curve: surges during handling, fall after vein ligation, with drugs marked.
- Catecholamine pathway to metanephrines (COMT).

> **Key points to remember:**
>
> - Right vein short into IVC; left vein into left renal vein.
> - StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
> - Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
> - Metanephrines first; never biopsy.
> - α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
> - Volume expansion prevents hypotension; watch glucose.
> - Arterial line before induction; avoid histamine releasers and ephedrine.
> - Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
> - After vein ligation: fluids, vasopressors, glucose.
> - One-third hereditary; SDHB most often metastatic.

LAQ 2 Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma

P2 · P4 **merged from 2**

Asked as: P4 LAQ "Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management" (asked as "Diagnosis and management for Adrenal Carcinoma", Summer 2018 Paper IV Q2 LAQ; "Approach to evaluation of adrenal mass/s", Summer 2022 Paper IV Q3f SAQ) · P2 SAQ "Algorithm for management of an incidentaloma" (Practice Set 3 Paper 2 Q3d)

Assumed meaning (P2): adrenal incidentaloma.

Definition and epidemiology

- **Adrenal incidentaloma:** mass **≥1 cm** found on imaging done for another reason.
- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**
- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **ACC syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

Differential diagnosis of an adrenal mass

- Non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC 4.7%; metastasis 2.5%; myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

Clinical features of ACC

- **Functional in 50–79% of adults** (90% of children):
- Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
- Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
- Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

Incidentaloma algorithm

Adrenal mass ≥1 cm ├─ IMAGING: unenhanced CT (computed tomography) │ ≤10 HU (Hounsfield units), homogeneous → benign (lipid-rich) adenoma │ (or MRI chemical-shift signal drop-out → lipid-rich adenoma) │ >10 HU → washout CT (15 min) or chemical-shift MRI │ APW >60% or RPW >40% → adenoma │ indeterminate → MDT (multidisciplinary team): repeat imaging 6-12 months, FDG-PET or surgery │ Macroscopic fat → myelolipoma ├─ HORMONES (all patients) │ 1-mg overnight DST (dexamethasone suppression test) │ plasma or urine metanephrines │ ARR (aldosterone-renin ratio) if hypertension or low potassium │ sex steroids only if ACC suspected └─ DECISION functioning, suspicious, >4 cm, or growing → adrenalectomy benign, non-functioning → discharge

Hormonal work-up (all lesions; ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

Axis / hormone	Tests
Glucocorticoid (cortisol)	**1-mg overnight DST** (or 24-h urine or late-night salivary cortisol): ≤1.8 µg/dL (50 nmol/L) excludes; 1.8–5.0 µg/dL = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 µg/dL autonomous; ACTH (adrenocorticotropic hormone)
Sex steroids and precursors	DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women (only if ACC suspected)
Mineralocorticoid (aldosterone)	**ARR** (aldosterone-to-renin ratio) if hypertension or hypokalaemia
Catecholamines	**Plasma free or 24-h urinary fractionated metanephrines**: must exclude phaeochromocytoma
Bilateral	17-hydroxyprogesterone (CAH); exclude adrenal insufficiency

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

Imaging

- Abbreviations: CT, HU, MRI (chemical-shift magnetic resonance imaging), MDT, FDG-PET (fluorodeoxyglucose positron emission tomography).
- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5% (another source: carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm). Benign imaging overrides size.
- **ACC size:** mean 10–12 cm; >90% exceed 5 cm.
- **CT in ACC:** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI in ACC:** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT:** ACC is avid, but PET cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

Management of incidentaloma (ESE–ENSAT 2023)

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features or suspicious imaging, >4 cm in fit patients, significant growth (≥5 mm and >20%). Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes and osteoporosis; surgery individualised.

- **Follow-up:** homogeneous **≤10 HU, non-functioning: benign, no further imaging whatever the size**; metanephrines may be omitted. Indeterminate non-resected: re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (17-hydroxyprogesterone); exclude adrenal insufficiency.

ACC pathology: Weiss score (≥3 of 9 = malignant)

Domain	Criteria
Nuclear / cytology	High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses
Architecture	≤25% clear cells; diffuse architecture; necrosis
Invasion	Venous; sinusoidal; capsular

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

ENSAT staging

Stage	Definition	5-year disease-specific survival (ENSAT 2009)
I	T1 (≤5 cm), N0 M0	~80%
II	T2 (>5 cm), N0 M0	~60%
III	T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1	~50%
IV	Distant metastasis only	~13%

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

Management of ACC

1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT [dichlorodiphenyltrichloroethane] derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
- Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
- Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.

- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit, so observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

3. Advanced or metastatic (stage IV)

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

Follow-up of ACC

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

Prognosis of ACC

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis.
- Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC).
- ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

Diagrams to draw

- The incidentaloma imaging and biochemistry algorithm above.
- ACC algorithm: 1. Adrenal mass: hormones + CT (HU, washout). 2. Suspected ACC: open R0 en bloc resection + LND (lymph node dissection). 3. Risk-based adjuvant mitotane or radiotherapy.

> **Key points to remember:**
> - ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
> - 1-mg DST: >1.8 µg/dL = MACS.
> - Exclude phaeochromocytoma before biopsy or surgery.
> - Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
> - Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
> - Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
> - Open, R0, intact capsule, en bloc + lymphadenectomy.
> - Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

LAQ 3 How will you evaluate and manage 38 yr male with uncontrolled hypertension

P2

Type: LAQ (25 marks)
Source: Practice Set 7 Paper 2, Q1 (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.

Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

Causes

Group	Causes	Clues
Renovascular	FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma	Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm
Renal parenchymal	CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney	Raised creatinine, proteinuria, haematuria
Obstructive uropathy	Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention	LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis
Primary aldosteronism (PA)	Adenoma (Conn), bilateral hyperplasia	Hypokalaemia (present in <40% only)
Phaeochromocytoma, paraganglioma	Sporadic or hereditary	Paroxysmal headache, sweating, palpitations
Cushing syndrome	Exogenous steroid, adrenal adenoma, pituitary	Central obesity, striae, diabetes
- **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation.		

Evaluation

Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

Step 2: History and examination

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.
- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

Step 3: Baseline tests and target-organ damage

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.
- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

Step 4: Targeted screening (method and interpretation)

Cause	Screen	Confirm or localise
PA	Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium	**Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm
Phaeochromocytoma	**Plasma free or 24-h urinary fractionated metanephrines**	CT or MRI (magnetic resonance imaging); ^{68}Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing
Cushing	**1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol	ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic
Renovascular	**Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5	CT or MR angiography; catheter angiography with pressure gradient if intervention planned
Parenchymal, obstructive	Ultrasound, urine tests	Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography

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Uncontrolled BP, age 38 → exclude pseudo-resistance → baseline tests + ultrasound KUB → ARR, metanephrines, dexamethasone test, renal Doppler ├─ positive → confirm → localise → treat cause └─ all negative → essential: optimise drugs
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Management

1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

2. Cause-specific

Cause	Treatment
PA, lateralised	**Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives
PA, bilateral or unfit	Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone
Phaeochromocytoma	**α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life
Adrenal Cushing	Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery
FMD	**Angioplasty alone** (cure common in the young)
Atherosclerotic RAS	Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function
Branch disease, aneurysm	Surgical bypass or ex vivo repair with autotransplantation
Reninoma	Partial nephrectomy (cures most)
Small non-functioning kidney (reflux, segmental hypoplasia)	**Nephrectomy** if renal vein renin lateralises (ratio >1.5)
Page kidney	Drain or decapsulate
Obstruction	**Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis
CKD	ACE inhibitor or ARB for proteinuria; scheduled diuretics

3. Follow-up: home BP, creatinine, potassium; duplex after angioplasty.

Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

Diagram to draw: stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Short answer questions (SAQ)

SAQ 1 Cushing syndrome and the dexamethasone suppression test

P2

Asked as:

- Cushing Syndrome — *Practice Set 2 Paper II Q3c (SAQ)*
- Dexamethasone suppression test — *Practice Set 5 Paper II Q3c (SAQ)*

Type: SAQ (10 marks)

Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

Causes

Group	Cause	Share
Exogenous	Glucocorticoids (oral, inhaled, topical)	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary adenoma (Cushing disease)	About 70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma)	10–15%
ACTH-independent	Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex)	About 10%

Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

Diagnosis

Step 1: exclude exogenous steroids (including creams and sprays) first. Step 2: confirm cortisol excess (two tests)

Test	Method	Abnormal
Overnight 1-mg DST (dexamethasone suppression test)	1 mg at 11 pm; 8 am cortisol	>1.8 µg/dL (50 nmol/L)
Late-night salivary cortisol	11 pm, twice	Raised
24-h UFC (urinary free cortisol)	Two collections	Raised; used in pregnancy

Step 3: locate the source

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

Dexamethasone suppression test

Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.

- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

Types

Test	Method	Abnormal
Overnight 1-mg LD-DST (low-dose DST; preferred screen)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	See below
48-hour 2-mg LD-DST	0.5 mg 6-hourly for 48 hours; cortisol at 48 hours	>1.8 µg/dL; improves accuracy
High-dose DST	8 mg overnight or 2 mg 6-hourly for 48 hours	Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress)

Interpreting the 1-mg test

Morning cortisol	Meaning
≤1.8 µg/dL (50 nmol/L)	Normal; >90% sensitive, rules out
1.8–5.0 µg/dL	Possible autonomous cortisol secretion
>5.0 µg/dL (140 nmol/L)	Autonomous secretion; >95% specific
- ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**.	
- If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]).	

Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).

Management

Cause	Treatment
Cushing disease	Trans-sphenoidal surgery (cure 60–80%); radiotherapy
Adrenal adenoma	**Laparoscopic adrenalectomy**
ACC	Open radical adrenalectomy, mitotane

Cause	Treatment
Ectopic	Resect source; otherwise bilateral adrenalectomy
AIMAH, PPNAD	Bilateral adrenalectomy

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

Perioperative care (urological relevance)

- **Contralateral adrenal is suppressed:** autonomous cortisol suppresses the opposite adrenal; adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- Give **perioperative steroid cover** and taper after adrenalectomy: **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

Diagram to draw: HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback and with ACTH-dependent and ACTH-independent sites.

> **Key points to remember:**
> - Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
> - 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
> - Oral contraceptives and enzyme inducers cause false positives.
> - ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
> - High-dose DST no longer routine.
> - Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

SAQ 2 MIBG scan

P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3c

Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

Isotopes

	^{123}I-MIBG	^{131}I-MIBG
Use	**Diagnostic** (preferred)	Older diagnostic; mainly **therapeutic**
Image quality	Better; allows SPECT-CT (single-photon emission computed tomography)	Poorer
Emission	Gamma	Beta (therapy) plus gamma

Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

Uses

1. Phaeochromocytoma and paraganglioma

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.
- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

2. Neuroblastoma

- ^{123}I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

3. Therapy (^{131}I-MIBG)

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.
- **FDG PET is now gold standard** for staging, and **^{68}Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

Diagram to draw: whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - ^{123}I diagnostic, ^{131}I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

SAQ 3 Adrenal Sx.

P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3d (preoperative phaeochromocytoma management also in

2020-Summer Paper 2, Q3c)

Assumed meaning: surgery of the adrenal gland (adrenalectomy).

Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

Approaches

Approach	Use
Laparoscopic transperitoneal (gold standard since 1992)	Familiar view, larger tumours
Retroperitoneoscopic (lateral or prone)	Prior abdominal surgery, small tumours
Robotic	Either route
Open (subcostal, chevron, thoracoabdominal, flank)	**ACC**, large or invasive tumours

Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.
- Conn: **hyperkalaemia** for weeks (high-salt diet).

Diagram to draw: right and left adrenal veins and relations.

> **Key points to remember:**
>
> - Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
> - Control the vein early; right vein short into IVC.
> - Open en bloc surgery for ACC.

Rapid Revision Sheet · Section 14

Phaeochromocytoma: adrenal anatomy, hormones and perioperative management (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma (LAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

Cushing syndrome and the dexamethasone suppression test (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

MIBG scan (SAQ)

- Noradrenaline analogue; block thyroid; stop labetalol.
- ^{123}I diagnostic, ^{131}I therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

Adrenal Sx. (SAQ)

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 15 · Renal, Nephrology & Transplant

Long answer questions (LAQ)

LAQ 1 Renovascular hypertension: pathophysiology, evaluation and treatment

P1 · P2 **merged from 2**

Asked as: P2 LAQ "Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment" (asked as "Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail", Summer 2022 Paper II Q2; "Discuss evaluation of & management of reno-vascular hypertension", Practice Set 2 Paper II Q1; "Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN.", Practice Set 4 Paper II Q2) · P1 LAQ "Pathophysiology of Renovascular hypertension" (Practice Set 8 Paper 1 Q2)

"Iscemic" is read as ischaemic nephropathy.

Anatomy of the renal vasculature

Renal arteries

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.
- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence of branching:** 1. Segmental 2. Lobar 3. Interlobar (columns of Bertin) 4. Arcuate (corticomedullary junction) 5. Interlobular 6. Afferent arteriole, glomerulus, efferent arteriole 7. Peritubular capillaries or vasa recta
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

Renal veins

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

	Right renal vein	Left renal vein
Length	2–4 cm	6–10 cm
Tributaries	None	Left adrenal, left gonadal, lumbar
Course	Direct to IVC	Between SMA and aorta (nutcracker syndrome)

	Right renal vein	Left renal vein
Anomalies	Accessory veins commoner	Circumaortic (2–16%); retroaortic

Relevance to RVH (renovascular hypertension): atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

Definitions

- **RVH:** hypertension caused by RAS (renal artery stenosis) that is relieved by revascularisation or nephrectomy. RAS is common in older (vascular) adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

Causes

	Atherosclerotic RAS	FMD
Share	Most (about 90%)	Most of the rest
Patient	Older, smokers, diabetic, vascular disease	Young women
Site	**Ostium and proximal third** (aortic plaque)	**Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type)
Course	Progresses to occlusion; renal function lost	Rarely occludes
Treatment	Medical first	**Balloon angioplasty without stent**; good BP (blood pressure) response

Also know: Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, **Page kidney** (perinephric compression, e.g. subcapsular haematoma), transplant RAS.

The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to: 1. low afferent arteriolar pressure (intrarenal baroreceptor); 2. low sodium chloride delivery to the **macula densa**; 3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising BP and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

Pathophysiology of RVH: Goldblatt models (basis of the tests)

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal. Loop: low perfusion pressure releases juxtaglomerular renin, which produces angiotensin II (vasoconstriction) and aldosterone (sodium retention).

	Two-kidney, one-clip (2K1C)	**One-kidney, one-clip (1K1C)**
Human equivalent	Unilateral RAS, normal opposite kidney	Bilateral RAS or stenosed solitary kidney
Sodium	Normal contralateral kidney excretes the extra sodium (**pressure natriuresis**)	No normal kidney: sodium and water retained
Renin	**High** from stenotic side; contralateral suppressed	Rises, then returns to normal as volume expands
Type	**Renin (angiotensin)-dependent**	**Volume-dependent**
ACE inhibitor effect	BP falls; captopril test positive	Less BP effect; may cause **acute renal failure / acute kidney injury**

Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less** (cure falls with duration). This explains poorer results in long-standing hypertension.

Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output** (P1: 20–25%), far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a significant translesional pressure gradient).
- Beyond this, the stenotic kidney keeps its GFR by angiotensin II-mediated **efferent constriction**. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence the **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

Evaluation: clinical clues (who to test)

- Onset <30 years (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.

- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidneys differing in size by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

Evaluation: basic tests

- Creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

Evaluation: imaging and functional tests

Test	Method and finding	Comment
Duplex Doppler US (ultrasound)	High PSV (peak systolic velocity), above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV**	First line, no contrast; operator dependent. **RI ≥0.8 (>0.8) means parenchymal disease and predicts poor response**
CT angiography	Stenosis, ostial plaque, beads	Best non-invasive anatomy; contrast nephropathy
MR (magnetic resonance) angiography	Gadolinium-enhanced	Caution if eGFR <30 mL/min
Captopril renography (DTPA, diethylenetriaminepentaacetic acid)	Day 1 after oral captopril: delayed uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of ≥10% = high probability RVH	Removing angiotensin II (efferent constriction) drops GFR in the stenosed kidney, proving renin-dependence; DTPA is cleared purely by filtration; use declining
Catheter angiography	Gold standard; **translesional (pressure) gradient** (systolic ≥20 or mean ≥10 mmHg = significant)	Done when intervention is planned
Renal vein renin	Ratio >1.5 lateralises	Before nephrectomy

Algorithm `Clinical clue -> Duplex US (or CTA/MRA if eGFR allows) -> no RAS: treat as essential hypertension -> RAS: FMD -> angioplasty Atherosclerotic -> optimal medical therapy -> high-risk features + viable kidney -> angiography + gradient -> stent`

Treatment 1: medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** controls renin-dependent BP; check creatinine and potassium at 1–2 weeks. Calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: most deaths in these patients are cardiovascular.

Treatment 2: landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting plus medical therapy versus medical therapy alone gave **no benefit** in renal function, BP, events or mortality; procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy gave **no reduction** in cardiovascular or renal events; only a small BP fall.

- Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

Treatment 3: indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

Treatment 4: endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion: angioplasty plus stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

Treatment 5: surgical revascularisation

- Indications: failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

Technique	Use
Aortorenal bypass (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene])	Standard
Transaortic endarterectomy	Bilateral ostial plaque
Splenorenal (left) / hepatorenal (right) bypass	Hostile (diseased) aorta
Ex vivo bench repair + autotransplantation	Branch FMD, aneurysms
Nephrectomy (partial or total)	Small non-functioning kidney with lateralising renin

Transplant RAS

- Raised velocity on Doppler; contrast imaging confirms; angioplasty or stent first, surgery if this fails.

Follow-up

- BP, creatinine, duplex at intervals for restenosis; cure is commoner in FMD than atherosclerosis.

Diagrams to draw

- 2K1C versus 1K1C with the renin–angiotensin loop (low pressure, renin, angiotensin II, aldosterone, hypertension).
- RAAS cascade from renin release to angiotensin II actions.
- Renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

Key points to remember:

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

LAQ 2 Allograft rejection and immunobiology of renal transplantation

P1 · P4 **merged from 2**

Asked as: P4 LAQ "Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs" (asked as "Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant", Winter 2017 Paper IV Q2; "Describe immunobiology of renal transplant and newer immunosuppressant drugs", Summer 2020 Paper IV Q1; "Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs", Practice Set 3 Paper IV Q1; "Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation", Summer 2015 Paper IV Q2) · P1 SAQ "Renal allograft rejection." (2020-Summer Paper 1 Q3d)

Definition and introduction

- Rejection is the recipient's immune attack on donor (allo) antigens, which the recipient's immune system recognises as foreign: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Immunosuppression has cut acute rejection from over 60% (pre-ciclosporin era) to **below 10%**. Slow, subclinical **chronic antibody-mediated injury** is now the main problem and the main cause of late graft loss.
- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

Transplant (target) antigens

Antigen	Features
HLA = MHC (major histocompatibility complex)	Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic
ABO blood group	On endothelium; incompatibility causes hyperacute rejection unless desensitised
Minor and non-HLA antigens	Rejection despite HLA identity

- **Sensitisation:** transfusion, pregnancy or previous transplant creates preformed anti-HLA antibodies.
- **Testing:** CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

Allorecognition (how T cells see the graft)

Pathway	Mechanism	Main role
Direct	Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells (donor antigen-presenting cells)	Very strong (high T-cell frequency); early acute TCMR (T-cell-mediated rejection)
Indirect	Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC**	Lifelong; helps B cells make DSA (donor-specific antibody), so alloantibody and chronic rejection
Semi-direct	Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it	Links both pathways

Three-signal model of T-cell activation

1. **Signal 1 (antigen):** TCR (T-cell receptor) binds MHC–peptide; **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells), which drives IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation): CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3:** IL-2 binds **CD25** (IL-2 receptor) and signals through **mTOR (mammalian target of rapamycin)**, moving the cell cycle from G1 to S; clonal proliferation follows, which needs de novo purine synthesis.

Effector mechanisms

- **Cellular:** CD4 Th1 cells (IL-2, IFN-γ [interferon-γ]) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme, causing **tubulitis and endarteritis (arteritis)**.
- **Humoral:** B cells with T-cell help become plasma cells that make **DSA**. DSA binds endothelium and fixes complement, leaving **C4d** on peritubular capillaries, with **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.
- **Regulation: Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

Types of rejection

Type	Timing	Mechanism and features
Hyperacute	Minutes–hours	Preformed ABO/HLA antibody; complement, thrombosis; graft turns blue; **prevented by crossmatch**
Accelerated acute	2–5 days	Memory (anamnestic) response in sensitised recipients
Acute TCMR	Days–months (mostly first 6 months)	Interstitial infiltrate, tubulitis, arteritis
Acute AMR	Any time	DSA, C4d, microvascular inflammation (capillaritis)
Chronic active AMR	Months–years	**Transplant glomerulopathy** (GBM [glomerular basement membrane] double contours), arterial intimal fibrosis; commonest (main) cause of late graft loss
Subclinical	—	Histological rejection, stable function

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant, previous rejection.

Diagnosis and monitoring

- **Clinical:** rising creatinine, oliguria; fever and graft tenderness are now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Monitoring: de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** has negative predictive value >90% and avoids unnecessary biopsy; gene-expression panels.
- **Biopsy is the gold standard;** Banff grades the findings and classifies TCMR and AMR.

Banff classification

Category	Key criteria
Borderline	Foci of tubulitis with mild interstitial inflammation
TCMR IA / IB	Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis
TCMR IIA / IIB	Mild-moderate (IIA) or severe (IIB) intimal arteritis
TCMR III	Transmural arteritis or fibrinoid necrosis
Active AMR	All three: (1) tissue injury (microvascular inflammation, arteritis, TMA [thrombotic microangiopathy]); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA**

Treatment of rejection

- **TCMR:** IV (intravenous) pulse methylprednisolone; **rATG (rabbit antithymocyte globulin) / ATG** for Banff II–III or steroid-resistant rejection.
- **AMR** (not standardised): steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

Marginal kidney donor: categories

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002: age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

2. KDPI (Kidney Donor Profile Index): percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

3. DCD (donation after circulatory death): Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of therapy	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either

Category	Description	Type
V	Euthanasia	Controlled

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

4. Others: AKI (acute kidney injury) donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en bloc); older or hypertensive living donors (KDIGO [Kidney Disease: Improving Global Outcomes] 2017 risk-based approach).

- **Absolute exclusions:** metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

Marginal kidney donor: assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program):** donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA matching. Informed consent is essential.
- **Ojo (JASN 2001):** ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI therapy.

Standard immunosuppression (current practice)

- **Induction** (over 90% receive it) plus **tacrolimus + mycophenolate** ± steroids (mycophenolate: GI [gastrointestinal] upset, teratogenic).
- Acute rejection is below 10%; chronic AMR is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

Newer immunosuppressive drugs

Drug	Mechanism (site)	Key points
Belatacept	**CTLA4-Ig**: binds CD80/86, blocks **CD28 costimulation (signal 2)**	FDA (Food and Drug Administration) 2011; IV 10 mg/kg induction, then **5 mg/kg 4-weekly** (monthly); CNI-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD [post-transplant lymphoproliferative disorder])
Extended-release tacrolimus	CNI: binds FKBP12 and inhibits calcineurin (**signal 1**), so less IL-2 transcription	Once daily, better adherence, steadier levels; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor

Drug	Mechanism (site)	Key points
Mycophenolate	Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis	Standard partner of tacrolimus
mTOR inhibitors (sirolimus, everolimus)	Bind FKBP12 and block mTOR (**signal 3**), arresting G1 to S	Not nephrotoxic; fewer CMV (cytomegalovirus)/BK infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced (low-dose) CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia
Basiliximab	Anti-CD25 (IL-2 receptor), non-depleting (**signal 3**)	Low-risk induction, **20 mg IV days 0 and 4**
rATG; **alemtuzumab**	T-cell depletion; anti-CD52	High-risk induction, DGF; infection (CMV), PTLD
Rituximab; bortezomib	Anti-CD20 (B cells); proteasome inhibitor (plasma cells)	AMR, desensitisation
Eculizumab	Anti-C5 (complement)	Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence
Imlifidase	Enzyme that **cleaves IgG**	**EMA (European Medicines Agency) 2020**: crossmatch-positive, highly sensitised patients
Anti-IL-6 therapy (tocilizumab, clazakizumab)	Blocks IL-6 pathway	Chronic active AMR

- **Also know:** anti-CD40 costimulation blockade is under trial. **Voclosporin** is approved for lupus nephritis, **not transplantation**.

Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

Diagrams to draw

- Three-signal T-cell activation (APC–T-cell synapse showing signals 1–3) with drug sites: CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis.
- Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

Key points to remember:

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

LAQ 3 Deceased donor kidney: selection, maintenance and retrieval techniques

P1 · P2 **merged from 2**

Asked as: P2 LAQ "Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval" (asked as "Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery", Summer 2021 Paper II Q2; "Describe procedure of identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt", Practice Set 3 Paper II Q1) · P1 SAQ "Kidney retrieval techniques for organ donation" (2023-Winter Paper 1 Q3c)

Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval. Marginal donor: see 2016-Summer Paper 3, Q3f; brain death certification: 2018-Summer Paper 4, Q3c; also answered in 2022-Summer Paper 4, Q3b.

Types of donor

- **Deceased (cadaver): DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).
- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- India: **THOA 1994**, amended 2011, **Rules 2014**, makes brain-stem death legal death; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

Identification and selection criteria

Potential donor: ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.

Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

Absolute contraindications

- Active or metastatic cancer, haematological malignancy, past melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

Standard versus marginal donor

Category	Definition
ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)	Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident)

Category	Definition
KDPI (Kidney Donor Profile Index)	Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival)
Paediatric <15 kg	En bloc transplant

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals.

Legal death and consent: certification of brain-stem death (THOA)

- **DBD preconditions:** apnoeic coma with a known irreversible cause; exclude drugs (sedatives, relaxants), hypothermia, metabolic causes and shock.
- **Board of four (4-member Board):** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**

Brainstem reflex (all absent)	Method	Cranial nerves
Pupillary	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI
Oculovestibular (caloric)	Ice-cold water into each ear (drum seen): no eye movement	VIII, III, VI
Cranial motor response	Supraorbital pressure: no grimace	V, VII
Gag and cough	Touch pharynx; suction catheter in trachea: no response	IX, X

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- **Consent:** next-of-kin (near relative) consent after a **mandatory request** by ICU staff; in medico-legal cases inform police.

Donor maintenance

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

Distribution (allocation)

- **NOTTO**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list, then region, then national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

Deceased multi-organ kidney retrieval (DBD)

1. **Checks:** death certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis laparotomy with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for and preserve aberrant (accessory) renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed (surface) ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc (last):** lift both with aorta, IVC, perinephric fat and ureters off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

DCD: after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

Back-bench (back-table) preparation

- Separate the kidneys: open the aorta posteriorly to see the renal ostia and divide it between them, keeping an aortic **Carrel patch** for each.
- Keep the IVC with the right kidney to lengthen the short right renal vein.
- Ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice at 4 °C; label.

Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

Technique	Features
Open (flank or mini-incision)	Complex anatomy, more often right side
Laparoscopic (Ratner, 1995)	Preferred; less pain, faster recovery
Hand-assisted	About two-thirds of US cases; tactile control
Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic	Selected centres

- **Sequence:** 1. Hilar dissection. 2. Ureter divided. 3. Artery then vein stapled or clipped. 4. Rapid extraction. 5. Cold flush.
- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), but not long-term graft loss.
- Keep cold ischaemia short: cold ischaemia >24 hours raises thrombosis risk.
- Normothermic perfusion is experimental.

Pitfalls

- Polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.

Diagram to draw

- Aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

> **Key points to remember:**
> - THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
> - Rule of 100s; vasopressin for hypotension and DI.
> - Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
> - Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
> - Left kidney preferred for living donation.
> - Machine perfusion reduces DGF.

LAQ 4 Vascular grafts and vascular surgery in urology

P1 · P2 · P3 **merged from 3**

Asked as: P3 LAQ "Vascular grafts and vascular surgery in urology" (asked as "Vascular Grafts in Urology", Summer 2021 Paper III Q2 LAQ; "Vascular Surgery in Urology", Winter 2024 Paper III Q3a SAQ) · P1 SAQ "Vascular grafts in urology" (Practice Set 5 Paper 1 Q3e) · P2 SAQ "Role of vascular grafts in urology" (2015-Summer Paper 2 Q3e)

Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. Tissue grafts (buccal mucosa) are not included.

Assumed meaning (vascular surgery): vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would fail or leave tension or narrowing.
- **Patch:** widens a partly resected wall. **Interposition graft:** replaces a segment. **Bypass:** routes flow around a lesion.

Classification of graft materials

Type	Examples	Advantages	Disadvantages
Autologous vein	Great saphenous, gonadal, superficial femoral, donor external iliac vein	Resists infection; suits small vessels and contaminated fields	Limited length; valves; harvest wound
Autologous artery	Internal iliac (hypogastric), **Carrel aortic patch**	Best for renal artery repair	Limited supply
Allograft (homograft)	Cadaveric vein or artery	Off the shelf; avoids xenograft	Availability; degeneration
Xenograft (biological)	**Bovine or porcine pericardium**	Pliable, strong patch, low infection	Patch only; cost
Synthetic	**PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron	Off-the-shelf, large calibre, any length	Infection needs removal; venous grafts thrombose

Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- **Choice of graft: autologous** vein or artery for small vessels and contaminated fields; **synthetic** (PTFE or Dacron) for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.

- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting, spatulated anastomosis with fine monofilament polypropylene (Prolene).
- **Sutures:** polypropylene (Prolene), double-armed. IVC or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- Antiplatelet or anticoagulant afterwards as indicated.
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

Application 1: renal transplantation

- Renal vein and artery end-to-side to external iliac vessels; Carrel patch.
- **Short right renal vein:** extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%):** Carrel patch; on the back table, conjoined side-to-side or end-to-side (bench) reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery:** mount the graft artery on a **Gore-Tex or bovine pericardial patch**.
- **Children under 15 kg:** aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%):** angioplasty or stent first; surgical interposition graft or bypass if it fails, which risks graft loss.

Application 2: RCC (renal cell carcinoma) with IVC tumour thrombus

- **Mayo levels 0–IV:** level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest.

Situation	Repair
Residual lumen ≥50%	Primary closure
Residual lumen <50%	**Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein
Circumferential wall invasion	**Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft
Infrarenal bland thrombus	Filter or caval interruption, sparing lumbar collaterals

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

Application 3: PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC:** narrowing <25%: primary repair; >25%: patch or interposition graft. **Ligation is tolerated** (slow occlusion builds collaterals).
- **Aorta:** never ligated; must be repaired or grafted. Small injury: repair or pericardial patch; resection: **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

Application 4: retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

Application 5: renovascular repair and autotransplantation

- Aortorenal bypass (saphenous vein, internal iliac artery, PTFE); splenorenal or hepatorenal bypass.
- Ex vivo bench repair of branch aneurysm or stenosis (or ureteric loss) with internal iliac artery or saphenous vein, then **autotransplantation** (see Q3e). Angioplasty or stent first for fibromuscular dysplasia.
- Traumatic renal artery injury: endovascular stent for the main renal artery; attempt open repair only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

Application 6: dialysis access

- Arteriovenous fistula first; **AV (arteriovenous) graft (PTFE)** when no native fistula is possible (poor veins): usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

Application 7: iatrogenic or traumatic vascular injury

- Interposition graft.
- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

Application 8: penile surgery

- Priapism: **Grayhack** saphenous vein–cavernosal shunt.
- Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.
- **Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational.

Application 9: endovascular

- Angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF after partial nephrectomy.

Complications

- **Thrombosis** (caval grafts 10–40%), anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic grafts usually need removal); graft–enteric or urinary fistula.
- **Ureteric:** transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

Diagrams to draw

- IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted.
- Mayo IVC thrombus levels with the matching vascular control.

- Bench arterial reconstruction.

> **Key points to remember:**
> - Proximal and distal control, heparin, flush, everting Prolene anastomosis.
> - Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
> - Mayo levels decide caval control and bypass.
> - IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
> - PC-RPLND: IVC may be ligated; aorta must be grafted.
> - Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
> - Bench surgery plus autotransplantation for complex renovascular lesions.

LAQ 5 Vascular anastomosis and vascular access for dialysis

P2 · P4 **merged from 2**

Asked as: P2 LAQ "Principles of vascular anastomosis and vascular access surgery for HD" (asked as "Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD", Practice Set 5 Paper II Q2 LAQ and Practice Set 12 Paper II Q3a SAQ) · P4 SAQ "Vascular access for dialysis." (2020-Summer Paper 4 Q3b)

Assumed meaning: HD = haemodialysis. Vascular access also answered in 2025-Winter Paper 2, Q3c.

Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).
- Haemodialysis needs a blood flow of about **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of vascular anastomosis

Principle	How and why
Exposure and control	Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps)
Anticoagulation	Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column
Atraumatic handling	Hold the adventitia only, never the intima; clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis
Healthy ends	Trim diseased ends; resect venous valve cusps at the suture line; prevents stenosis and leak
No tension, twist or kink	Adequate length; trial the final position before suturing
Intima-to-intima eversion	Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic
Wide opening	Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis (stricture)
Suture	**Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0
Needle direction	Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down, not lifted into a flap
Continuous or interrupted	Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth
Technique	**Carrel triangulation** (three stay sutures keep the back wall clear); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint

Principle	How and why
Flush and release	Before the last knot, briefly release clamps to flush air, clot and debris; leave slack ("growth factor") to avoid purse-stringing; release venous/distal clamp first, then arterial
Confirm	Haemostasis, pulse, thrill, Doppler

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

Options for vascular access surgery for HD

Option	Construction / description	Ready	Comment
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein	At once	AKI (acute kidney injury) and emergencies; acute dialysis, days only
2. Tunnelled cuffed CVC	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula)	Artery joined to a superficial vein, usually vein end to artery side	6–8 weeks (matures)	**First choice:** best patency, fewest infections
a. Snuffbox or **radiocephalic (Brescia–Cimino, 1966)**	Radial artery to cephalic vein at wrist	6–8 weeks	Most distal; preserves proximal sites; higher failure to mature (most fail to mature)
b. Brachiocephalic	Brachial artery to cephalic vein at elbow	6–8 weeks	Higher flow; more steal
c. Brachiobasilic transposition	Deep basilic vein mobilised and moved superficially	Longer, often two stages	When the cephalic vein is unusable
4. AVG (arteriovenous graft)	PTFE loop or straight: forearm loop, upper-arm or thigh	2–3 weeks (usable)	Poor veins; more stenosis (venous anastomosis), thrombosis, infection
5. HeRO (Haemodialysis Reliable Outflow) graft	Graft outflow bypasses occluded central veins into the right atrium		Central vein occlusion
6. Endovascular AVF (WavelinQ, Ellipsys)	Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels		No incision; selected anatomy

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "ESKD Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.

- **Vein preservation:** preserve arm veins; no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications of access

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty (PTA), ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then treat the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft, remove catheter

- **Also know:** venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

Diagram to draw

- Radiocephalic end-to-side fistula at the wrist with Carrel triangulation stay sutures.

> **Key points to remember:**
> - Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
> - AVF first, distal, non-dominant arm, about 6 months ahead.
> - Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal: banding or DRIL.

LAQ 6 Describe in details basic patho-physiology of urine production and transport

P1

Type: LAQ (25 marks)
Source: 2015-Summer Paper 1, Q1

Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
- **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
- **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

D. Tubular handling

Segment	Key transporter	Handling	Drug/disease link
Proximal tubule	Na^+/H^+ exchanger, SGLT2 (sodium–glucose cotransporter 2)	~**65%** Na^+ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs	SGLT2 inhibitors
Thin descending limb	Aquaporin-1	Water out, solute stays: fluid concentrates	–
Thick ascending limb	NKCC2 (Na^+-K^+-$2Cl^-$)	~**25%** NaCl; water-impermeable "diluting segment"	Loop diuretics

Segment	Key transporter	Handling	Drug/disease link
Distal convoluted tubule	NCC (Na^+-Cl^- cotransporter)	5–7% Na^+; Ca^{2+} (parathyroid hormone)	Thiazides lower urinary Ca^{2+}
Collecting duct	ENaC (epithelial Na^+ channel), AQP2	Principal cells: Na^+/K^+ (aldosterone), water (ADH); intercalated cells: acid	Amiloride, spironolactone; distal RTA (renal tubular acidosis)

E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

Hormone	Site	Effect
Angiotensin II, aldosterone	Efferent arteriole, proximal tubule; collecting duct	Na^+ retention, K^+/H^+ loss, GFR maintained
Atrial natriuretic peptide	Afferent arteriole, collecting duct	Natriuresis, lowers renin
Parathyroid hormone	Proximal and distal tubule	Phosphaturia, Ca^{2+} retention

F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 cmH_2O. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 cmH_2O stops efflux.

Factor	Effect on transport
Obstruction	Acute: stronger contractions; chronic: dilatation, lost coaptation
Infection	Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction)
Pregnancy	Right > left dilatation: compression at pelvic brim, progesterone
Drugs	α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic

G. Bladder storage and voiding

- **Storage** below ~10 cmH_2O: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

H. Pathophysiology of obstruction (unilateral: triphasic)

`Phase 1 (0–1.5 h): RBF up, pressure up (afferent dilatation: prostaglandins, nitric oxide) Phase 2 (1.5–5 h): RBF down, pressure up (efferent constriction) Phase 3 (>5 h): RBF down, pressure down (afferent constriction: angiotensin II, thromboxane A2, endothelin)`

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and Na^+ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.
- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 cmH_2O = obstructed); tolvaptan in polycystic disease.

Diagram to draw: Nephron with transporters; triphasic graph.

> **Key points to remember:**
> - RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
> - Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
> - Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

LAQ 7 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL

P1

Asked as:

- Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries — *Winter 2017 Paper I Q1 (LAQ)*
- Anatomy of renal collecting system and its relevance to PCNL — *Practice Set 1 Paper I Q3d (SAQ)* (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)
- Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. — *Summer 2022 Paper I Q1 (LAQ)* (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.
- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

Surgical anatomy relevant to PCNL: position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

Anatomy of the calyces (collecting system)

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**; it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** (2–3 papillae) are commonest at (cluster at) the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region (hilar ones) by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane (posteriorly), so **posterior calyces point posterolaterally** (towards the prone surgeon). On a pyelogram anterior and posterior calyces (rows) overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces (does not link upper and lower calyces). Suspect it when one pole cannot be reached from a puncture.

Sampaio classification (Sampaio types; 140 resin endocasts)

Type	Frequency	Midzone (hilar) drainage
A-I	45%	Midzone calyces drain into the superior or inferior calyceal group (via a polar group)
A-II	17.2%	**Crossed calyces**: one to the superior and one to the inferior group (to both groups)
B-I	21.4%	Independent major calyceal group (major calyx) from the midzone
B-II	16.4%	1–4 minor calyces enter the pelvis directly (straight into the pelvis)

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram (pyelography); narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the **columns of Bertin**, hugging the anterior and posterior faces of the infundibula (beside the infundibula).
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to (behind) the lateral border, along the posterior calyces. Posterior calyces along it are the target.

Puncture route / site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%); posterior segmental artery
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Infundibular artery (present) in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

Relations (the hazards of access)

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

Structure	Relation	Practical point
Pleura, lung	Pleura reaches the 12th rib; lung the 10th intercostal space	Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration
Liver, spleen	Over the upper poles; descend in inspiration	Risk with supracostal access, inspiration, organomegaly

Structure	Relation	Practical point
Colon	Retrorenal colon, mainly at the lower pole, commoner prone and on the left	Review CT; ultrasound shows bowel

Applications in endourology

1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured (entered) **through the papilla and fornix** along the calyx axis, never into (through) the infundibulum or pelvis; identified by injected air, C-arm rotation or ultrasound. This passes through a pyramid and avoids interlobar arteries.
- **Upper-pole posterior access** gives the straightest route (best reach) to the pelvis, UPJ and most calyces (staghorn; staghorn branches). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks (retrorenal) colon (left) and reaches the UPJ poorly (poor UPJ reach).
- Identifying (identify) posterior calyces: rainbowing the C-arm towards the surgeon shortens them (they shorten when the C-arm rotates towards the surgeon); injected air fills non-dependent (posterior, when prone) calyces (rises posteriorly when prone); ultrasound also shows bowel and pleura.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).
- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

4. Infundibulotomy and calyceal diverticulum

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

5. Open/anatrophic nephrolithotomy: incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); transplant kidneys; megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

Mini-PCNL: definition

Term	Tract size
Standard	24–34 Fr
Mini	15–22 Fr
Ultra-mini	11–14 Fr
Micro-perc	<10 Fr

- First mini-perc: Jackman, 1998 (13 Fr).

Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.
4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.
8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

Complications (grade by Clavien–Dindo)

Type	Points
Bleeding	Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation
Infection	Fever, SIRS (systemic inflammatory response syndrome), urosepsis
Thoracic	Pneumothorax, hydrothorax: chest drain

Type	Points
Visceral	Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum
Collecting system	Perforation, extravasation, later infundibular stenosis; residual stones
Others	Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death

Diagram to draw: transverse section of kidney showing anterior and posterior calyces (posterior calyx), Brödel line and the needle through the posterior calyceal fornix.

> **Key points to remember:**
>
> - Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
> - Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
> - Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
> - Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
> - Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
> - High intrarenal pressure in mini-PCNL causes fever and sepsis.

LAQ 8 Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management)

P1

Asked as:

- Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria — *Summer 2018 Paper I Q1 (LAQ)*
- Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. — *Practice Set 8 Paper I Q1 (LAQ)*
- Management of chyluria — *Practice Set 10 Paper I Q1 (LAQ)*

Type: LAQ (25 marks)
(also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

General plan of lymphatic drainage

- Lymph flows **cephalad and right → left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left (the two sides). Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.
- **Lymphatic anatomy (summary):** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.

Drainage by organ

Organ	Primary nodes	Notes
Left kidney	Left lateral para-aortic	Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural
Right kidney	Right paracaval, interaortocaval	May cross to left para-aortic
Ureter	Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac	Plexus in muscle and adventitia

Organ	Primary nodes	Notes
Bladder	Mainly **external iliac**; also internal iliac, obturator	Begins in lamina propria; bilateral cross-drainage
Prostate	**Obturator and internal iliac**; presacral	Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal)
Testis	Left: left para-aortic; right: interaortocaval, then right paracaval	Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac
Penis	Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic	Shaft lymphatics converge dorsally
Scrotum	Ipsilateral superficial inguinal	Does not cross the raphe
Urethra (male)	Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator	Female: proximal → pelvic; distal → inguinal

Lymphatic drainage of the urinary tract (intrinsic lymphatics and regional nodes)

Organ	Intrinsic lymphatics	Regional nodes
Kidney	Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein	**Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic)
Ureter	Plexuses in the muscle and adventitia	Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac
Bladder	Begins in the lamina propria; a few perivesical nodes	Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides
Prostate	Periprostatic plexus	Obturator and internal iliac; presacral; external iliac
Urethra	Submucosal plexus	Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, giving milky urine (making it milky; urine looks milky). Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices / a renal calyceal fornix (**pyelolymphatic fistula**).

Etiology

Parasitic (commonest; India, China)	Non-parasitic
Wuchereria bancrofti (lymphatic filariasis; *Culex* mosquito)	Congenital lymphangiectasia, renal lymphangioma

Parasitic (commonest; India, China)	Non-parasitic
Brugia (lymphatic filariasis that spares the genitals; rarely urological)	Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection))
	Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy

Pathogenesis (obstructive theory)

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- **Obstructive (reflux) theory:** `Adult worms + host immune reaction -> lymphangitis and fibrosis -> obstruction between intestinal trunks/cisterna chyli and thoracic duct -> dilatation of retroperitoneal lymphatics, valve incompetence -> reflux of chyle into renal hilar and peripelvic lymphatics -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix -> pyelolymphatic fistula -> chyluria`
- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest. So every treatment is judged against natural remission.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

Clinical features

- Milky urine, worse after fatty meals; **chylous clots** (retention, colic); **haematochyluria**.
- Loss of fat and protein → weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

Grade	Features
Mild	Intermittent; no systemic effect
Moderate	Persistent; clots

Grade	Features
Severe	Haematochyluria, clot retention, weight loss, hypoalbuminaemia

Evaluation (before treatment)

1. Confirm chyle

- **Ether (chloroform) test**: milky urine shaken with ether clears because fat dissolves. **Urine triglycerides** (and protein) positive/raised; Sudan III stains fat globules.
- Distinguish from / exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

2. Filarial status (filarial tests)

- Night blood smear (often negative at this stage / late), **circulating filarial antigen** (card) test, eosinophilia; PCR (polymerase chain reaction).

3. Lateralise the side and localise the fistula

- **Cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice (one, sometimes both).
- **Retrograde pyelography** shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR (magnetic resonance) lymphangiography or CT.

4. Baseline: serum albumin, haemoglobin, renal function; urine culture before instillation.

Management (outline / algorithm)

`Mild → diet + DEC Persistent / moderate → add sclerotherapy (can repeat) Severe or failed sclerotherapy → renal hilar lymphatic disconnection Mild -> low-fat, high-protein diet + MCT oil + DEC Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine) Severe or failed -> renal hilar lymphatic disconnection (open, laparoscopic, robotic) Confirm chyle -> grade severity Mild -> diet + antifilarial drugs; observe Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection`

1. Conservative (all patients)

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids.
- **Why MCT:** MCTs (medium-chain fats) are absorbed straight/directly into the portal blood, not into lacteals, so chyle formation falls (less chyle is formed).
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae (another source: 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective). It has only modest effect on adults.
- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont; anti-*Wolbachia*) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)

- Ureteric catheter on the affected (leaking) side, head-down position. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial success/response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every/all hilar lymphatics. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy ± optional **omental wrap** around the hilum as a barrier. 5. **On-table cystoscopy** confirms clear urine/efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy (now) obsolete; prevent by clipping lymphatics during renal and nodal surgery.

4. Follow-up and prevention

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

Diagram to draw: Retroperitoneal lymphatics with cisterna chyli, obstruction (a block above it / obstructed thoracic duct), dilated hilar lymphatics, and reflux to a pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
> - Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
> - Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
> - Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
> - Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
> - Retroperitoneoscopic disconnection beats open.

LAQ 9 Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD)

P2

Asked as:

- Discuss principles of hemodialysis and vasular access in details. — *Winter 2017 Paper II Q1 (LAQ)*
- Vascular access for hemodialysis — *Winter 2025 Paper II Q3c (SAQ)*
- Hemodialysis access. — *Practice Set 10 Paper II Q3c (SAQ)*
- Access techniques in dialysis — *Practice Set 12 Paper II Q3f (SAQ)*
- Discuss techniques of dialysis — *Practice Set 5 Paper II Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(Winter 2025 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b) (Practice Set 10 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c) (Practice Set 12 Paper II Q3f: CAPD also answered in Practice Set 1 Paper 2, Q3a) (Practice Set 5 Paper II Q3e: HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

Assumed meaning (Winter 2017): "vasular" is a typo for "vascular" access for haemodialysis. Assumed meaning (Practice Set 12): access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a. Assumed meaning (Practice Set 5): principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.

Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about **250–350 mL/min** of blood flow.
- Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Principles of haemodialysis (physical principles)

Process	Driving force	What it removes
Diffusion	Concentration gradient across the membrane	Small solutes (urea, creatinine, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute is dragged with water	Middle molecules (e.g. β2-microglobulin)
Ultrafiltration	Transmembrane (hydrostatic) pressure (HD) or osmotic (PD) gradient	Excess water, down to "dry weight"

- **Countercurrent flow:** dialysate runs opposite to blood (**countercurrent**), keeping (to keep) a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** blood (250–350 mL/min) passes through hollow fibres (hollow-fibre) of **synthetic membrane**; high-flux membranes clear more middle molecules.

- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.
- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h (hours) three times (3 times) a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).

Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 (Kt/V ≥1.2) per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).
- **Access-related:** access infection, thrombosis, steal.

Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously (see PD access below).
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).

- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

HD versus PD

	HD	PD
Setting	Centre or home	Home, continuous
Haemodynamics	Rapid shifts	Gentle; preserves residual function
Favoured in	Abdominal surgery, failed PD	Children, cardiac instability, no vascular access

Types of vascular access

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino, 1966)	Radial artery to cephalic vein at the wrist, end-to-side (now usually vein end to artery side)	≥6–8 weeks	**First choice**: best patency, fewest infections; highest failure to mature
Brachiocephalic AVF	Brachial artery to cephalic vein at the elbow	≥6–8 weeks	Higher flow; more steal
Brachiobasilic (transposition) AVF	Deep basilic vein transposed superficially	Longer	When the cephalic vein is unusable
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; highest infection and central stenosis

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first** (US **Fistula First**, 2003). **KDOQI 2019** asks for (favours) an individual "ESKD Life-Plan": the right access for the right patient at the right time.
- **Timing:** refer at eGFR <30; create an AVF at least (about) **6 months** before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm (that arm); avoid subclavian catheters (lines).
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by (at) 6 weeks.

Access complications

Complication	Key point	Treatment
Failure to mature	Juxta-anastomotic stenosis; side veins	Angioplasty; ligate (side) branches
Stenosis	Neointimal hyperplasia; AVG at venous anastomosis	PTA (percutaneous transluminal angioplasty), stent graft, revision

Complication	Key point	Treatment
Thrombosis	Usually on an underlying stenosis	Thrombectomy, then treat (fix) the stenosis
Steal syndrome	Hand ischaemia; diabetes, brachial-based access	Banding, DRIL (distal revascularisation–interval ligation), ligation
Aneurysm	Repeated needling at one site	Repair if skin is thin
Infection	CVC > AVG > AVF	Antibiotics; excise graft; remove catheter

- **Also know:** brachiobasilic transposition AVF; venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

Peritoneal dialysis access

Catheter: Tenckhoff silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

Insertion techniques

Technique	Steps	Comment
Open (mini-laparotomy)	Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site	Simple; general or local anaesthesia
Laparoscopic (advanced)	Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed	Best function; fixes causes of outflow failure
Percutaneous (Seldinger)	Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy	Bedside; avoid after major abdominal surgery
Peritoneoscopic	Small-scope guided placement	Variant of percutaneous
- **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes.		
- **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation.		
- **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges.		

Haemodialysis catheter access (brief, insertion)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

Recent advances

- **Endovascular AVF** (percutaneous creation: WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

Diagram to draw: dialyser with countercurrent blood and dialysate (beside a PD catheter in the pelvis), and a radiocephalic end-to-side fistula at the wrist; Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

> **Key points to remember:**
>
> - Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
> - HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
> - AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
> - Tenckhoff: two cuffs, pelvic tip, downward exit.
> - Laparoscopic placement with omentopexy gives best function.
> - Avoid subclavian lines; AVF first for long-term HD.
> - CRRT for unstable AKI; PD preferred in children.

LAQ 10 Acute renal failure (AKI) and acute tubular necrosis

P2

Asked as:

- Etiology, pathogenesis and management of acute renal failure. — *Winter 2024 Paper II Q2 (LAQ)* (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)
- Acute tubular necrosis. — *Summer 2022 Paper II Q3c (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Acute renal failure is now called AKI (acute kidney injury).

Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI (per KDIGO) = rise in serum creatinine **or** fall in urine output; either alone is enough.

Stage	Serum creatinine	Urine output
1	Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days	<0.5 mL/kg/h for 6 h
2	×2 baseline	<0.5 mL/kg/h for 12 h
3	×3, or ≥4 mg/dL, or RRT (renal replacement therapy)	<0.3 mL/kg/h for 24 h, or anuria 12 h

- One-fifth to one-third of inpatients; mortality about 25%, about 50% (about half) if RRT is needed.
- **ATN definition:** ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

Etiology

Type	Causes
Prerenal	Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors
Intrinsic	**ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli)
Postrenal	BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

Causes of ATN

Ischaemic	Toxic
Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)**	**Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains

Pathogenesis (pathophysiology)

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).

- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**
- **Why tubules:** the medulla works near hypoxia (medullary oxygen tension is only 10–20 mmHg), while the **S3 proximal tubule (segment)** and **medullary thick ascending limb** consume much oxygen and are most vulnerable.
- **Loss of polarity:** Na^+/K^+-ATPase loses its basolateral position (moves off the basolateral membrane) → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole (afferent constriction) → GFR falls (protective).
- Cells die (mostly apoptosis) and slough (dead cells slough) → **muddy-brown casts** block (obstruct) tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.
- Phases (clinical course): initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.
- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6 weeks).

Evaluation (diagnosis: ATN versus prerenal)

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

Index	Prerenal	ATN
Urine sodium	<20 mmol/L	>20 mmol/L
FeNa (fractional excretion of sodium)	<1%	**>1%**
FeUrea (on diuretics)	<35%	>35%
Specific gravity	>1.015	<1.010
Sediment	Bland	**Muddy-brown granular casts**, tubular cells
Fluid challenge (response to fluids)	Quick (rapid) recovery	Lag of days to weeks

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags (fails).

Management

General

- Treat (remove) the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers) (RAAS (renin–angiotensin–aldosterone system) blockers); adjust drug doses.

- Isotonic crystalloid (fluids) for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast (pre-contrast hydration).
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

Complications

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics (only for volume overload) do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

Dialysis indications (AEIOU), refractory to medical therapy

	Indication
A	Acidosis
E	Electrolytes, mainly hyperkalaemia
I	Intoxications / ingestions (lithium, salicylate, alcohols)
O	Overload with respiratory distress
U	Uraemia: encephalopathy, pericarditis, bleeding

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

Post-obstructive diuresis

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has no role.

Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD. ATN recovery takes days to weeks.
- **Also know:** (early) biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

Diagram to draw: tubular cell losing polarity → sloughing → casts and back-leak.

Key points to remember:

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

LAQ 11 Discuss the role of tissue typing and cross match in live and cadaveric

P3

Type: LAQ (25 marks)
Source: 2018-Summer Paper 3, Q2

The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".

Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

	Class I	**Class II**
Loci	HLA-A, -B, -C	HLA-DR, -DQ, -DP
Expression	All nucleated cells	Antigen-presenting cells, endothelium
Presents to	CD8 (cytotoxic) T cells	CD4 (helper) T cells

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

Tissue typing methods

Method	**Principle**	**Note**
Serological (microlymphocytotoxicity)	Lymphocytes + typed antisera + complement; dead cells take up dye	Low resolution; largely replaced
Molecular (DNA-based)	PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing)	High (allele-level) resolution; now standard

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.

- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

Crossmatch

Type	Method	Meaning if positive
CDC (complement-dependent cytotoxicity)	Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity	**Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969)
Flow cytometry crossmatch	Antibody binding on donor cells detected by fluorescent anti-IgG	More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange
Virtual crossmatch	Compare recipient's bead-defined antibodies with donor HLA type	Speeds deceased-donor allocation; confirm by cell-based crossmatch

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.
- Crossmatch uses a **recent serum** plus historical peak sera.

Living versus deceased donor

	Living donor	Deceased (cadaveric) donor
Typing	ABO and HLA early in donor work-up; choose best of several donors	ABO and HLA at donor referral
Crossmatch	At work-up and a **final crossmatch shortly before surgery**	At the time of offer, often virtual first; done against the list rapidly (cold ischaemia)
Role of matching	Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995)	Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index)
If incompatible	**Paired (swap) exchange**, desensitisation, ABO-incompatible protocol	Patient waits; acceptable-mismatch programmes

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

Diagram to draw: CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Short answer questions (SAQ)

SAQ 1 Brain-stem death: pathophysiology, guidelines and declaration

P1 · P2 · P4 **merged from 3**

Asked as: P2 SAQ "Brainstem death: biology and clinical implications" (asked as "Brain stem Death: Biology and clinical implications", Winter 2024 Paper II Q3d; "Brainstem death", Practice Set 13 Paper II Q3c) · P1 SAQ "Patho-physiology of Brain stem death" (2015-Summer Paper 1 Q3d) · P4 SAQ "Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee)" (asked as "Brain dead committee", Summer 2018 Paper IV Q3c; "Guidelines for declaration of brain stem death", Summer 2022 Paper IV Q3b)

Assumed meaning (P4): the Board of Medical Experts that certifies brain-stem death under Indian law.

Definition and legal basis

- **Brain-stem death:** irreversible loss of the capacity for consciousness and of the capacity to breathe, from permanent brainstem failure; equivalent to death of the person (legally death).
- **India: THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).

- Donation after brain death (DBD) keeps the circulation until procurement, so warm ischaemia is minimal.

Biology (pathophysiology)

Sequence

1. Brain injury causes oedema and raised ICP (intracranial pressure).
2. Cerebral perfusion falls, leading to herniation ("coning").
3. Pons: Cushing reflex (hypertension, bradycardia).
4. Medulla: catecholamine storm (hypertension, tachyarrhythmia).
5. Loss of sympathetic tone: vasodilatation, hypotension.

System	Consequence
Heart and vessels	Storm causes myocardial injury and arrhythmia, then **hypotension**
Lungs	Neurogenic pulmonary oedema
Pituitary	**Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol
Hypothalamus	**Hypothermia** (poikilothermia)
Blood	DIC (disseminated intravascular coagulation) from brain thromboplastin
Whole body	Cytokine injury to organs; hyperglycaemia

Composition of the Board of Medical Experts (India; Rules 2014)

Member	Role
1. Registered medical practitioner **in charge of the hospital**	Hospital authority
2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority	Independent check
3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist)	Expert
4. Medical practitioner **treating the patient**	Clinical history

- **No member may belong to the transplant team.**

Certification: preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude drugs (**sedatives, relaxants**), hypothermia (35–36 °C by protocol), metabolic or endocrine causes, shock.

Certification: brainstem reflexes (all must be absent)

Brainstem reflex	Method	Cranial nerves
Pupillary light reflex	Bright light: pupils fixed, no reaction	II, III
Corneal	Touch cornea: no blink	V, VII
Oculocephalic (doll's eye)	Turn head: eyes move with the head	VIII, III, VI

Brainstem reflex	Method	Cranial nerves
Oculovestibular (caloric)	~50 mL ice-cold water into each ear (drum visible): no eye movement	VIII, III, VI
Motor response in cranial nerve area	Central painful stimulus (supraorbital pressure): no facial grimace	V, VII
Gag and cough	Touch pharynx; pass suction catheter down trachea: no response	IX, X

Certification: apnoea test and timing

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; watch for breathing effort; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline) is a positive test. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** are not mandatory; used when clinical examination cannot be completed (e.g. facial trauma): EEG (electroencephalogram), cerebral angiography, perfusion scan, transcranial Doppler.

Clinical implications: linked duties after declaration

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate; a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

Clinical implications: donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, PaO_2 (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (pressor, also treats DI); thyroid hormone, methylprednisolone, insulin; warming; protective ventilation.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

Diagrams to draw

- Herniation sequence leading to storm, DI, hypothermia, DIC.
- Certification flowchart: 1. Coma. 2. Preconditions. 3. Test 1. 4. 6 hours. 5. Test 2. 6. Form 10. 7. Consent. 8. NOTTO.

Key points to remember:

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

SAQ 2 GFR estimation in prospective renal donor

P1 · P2 **merged from 2**

Asked as: P1 SAQ "GFR estimation in a prospective renal donor and creatinine clearance" (asked as "Describe methods to estimate GFR in a prospective renal donor", Summer 2019 Paper I Q3a; "Creatinine clearance", Practice Set 4 Paper I Q3d) · P2 SAQ "GFR estimation: clinical assessment and in the prospective transplant donor" (asked as "Methods for GFR estimation in prospective tx donor", Practice Set 9 Paper II Q3a; "GFR clinical assessment", Winter 2023 Paper II Q3a)

Assumed meaning: tx = transplant (prospective living kidney donor).

Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 m^2. It is the standard index of kidney function.
- The ideal marker is freely filtered and neither secreted nor reabsorbed.

Why it matters in the donor

- The donor loses about half the nephron mass; function settles at 60–80% of the pre-donation level.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR thresholds and asks for **measurement, not just estimation**, at a decision point: GFR is assessed by at least two methods, **including a measured one**.

Method 1: serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle and is freely filtered (not reabsorbed), but generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age. It is also secreted, so it **overestimates GFR**, increasingly as function falls. Trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term**; currently recommended; used for staging. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

Method 2: cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, then fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone; it is the most accurate estimate.

Method 3: measured 24-hour creatinine clearance

- **Definition:** creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates GFR because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 m² in young men, slightly lower in women; falls with age.
- **CrCl = (U × V) / P**: U = urine creatinine, V = urine flow (mL/min), P = plasma (serum) creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

Limitations of creatinine clearance

- **Tubular secretion** of creatinine is small in health but rises as function falls, so CrCl **overestimates GFR** in CKD (chronic kidney disease).
- Collection errors are common. Creatinine generation depends on muscle mass, diet and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.
- Invalid in AKI (acute kidney injury).
- **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

Method 4: mGFR (measured GFR), the reference standard

Marker	Method / comment
Inulin	Urinary clearance; gold standard, research only
Iohexol	Plasma clearance; safe and accurate (another source: plasma or urinary clearance)
Iothalamate	Urinary or plasma clearance
^{51}Cr-EDTA (ethylenediaminetetraacetic acid)	Plasma clearance
^{99m}Tc-DTPA (diethylenetriaminepentaacetic acid)	Plasma sampling, or gamma camera (Gates method); purely filtered; also gives **split function**

Method 5: split function

- DTPA renogram gives each kidney's share (differential function); MAG3 or DMSA (dimercaptosuccinic acid) are alternatives.
- A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

KDIGO 2017 donor decision

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys** (creatinine–cystatin C eGFR).

GFR (mL/min/1.73 m²)	Decision
≥90	Accept

GFR (mL/min/1.73 m²)	Decision
60–89	Individualise (age, lifetime risk)
<60	Decline

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

Clinical uses of GFR and creatinine clearance

- **CKD staging (KDIGO):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m², always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor evaluation:** KDIGO 2017 table above; confirm with measured GFR.
- **Urology:** drug and contrast dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit); continent diversion (>35 mL/min); partial versus radical nephrectomy.
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable: measure creatinine clearance.
- **Acute change:** eGFR is invalid when function is unstable; AKI uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

Diagrams to draw

- Plasma tracer clearance curve after a single injection.
- Nephron showing filtration plus a small secreted fraction of creatinine.

> **Key points to remember:**
> - Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
> - Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
> - Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
> - KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
> - Better kidney stays with the donor.
> - CrCl = UV/P; check completeness by creatinine excreted per kg.
> - Secretion makes CrCl overestimate GFR as function falls.
> - Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

SAQ 3 Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens

P1 · P2 · P4 **merged from 3**

Asked as: P1 SAQ "Immunosuppressants in renal transplantation (classification, protocols, ciclosporin)" (asked as "Immunosuppressants", Summer 2021 Paper I Q3f; "Classify immunosuppresants. Describe few common protocols", Practice Set 3 Paper I Q3b; "Cyclosporins", Practice Set 4 Paper I Q3f) · P2 SAQ "Pre transplant Immunosupression treatment options" (2023-Winter Paper 2 Q3e) · P4 SAQ "Steroid-free immunosuppression in renal transplantation" (Practice Set 3 Paper 4 Q3f)

Taken as immunosuppressant drugs in renal transplantation; "Cyclosporins" taken as ciclosporin (cyclosporin A); "pre-transplant" taken as induction immunosuppression and

desensitisation given before or at transplantation. Also answered in 2020-Summer Paper 4, Q1; tacrolimus in 2017-Winter Paper 3, Q3c.

Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binds donor antigen; calcineurin then drives IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor signalling through mTOR [mammalian target of rapamycin], driving proliferation). Each drug blocks a step.
- Rejection risk is highest in the first weeks.

Phases

- **Induction** (peri-transplant intense immunosuppression), **maintenance** (lifelong), **rescue** (treatment of rejection).
- **Desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.

Classification

Class	Drug(s)	Mechanism / action	Main toxicity
CNI (calcineurin inhibitors)	**Tacrolimus**, ciclosporin	Block calcineurin (signal 1)	**Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism
Antimetabolites (antiproliferatives)	**Mycophenolate**, azathioprine	Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase])	Diarrhoea, leucopenia
mTOR inhibitors	Sirolimus, everolimus	Block signal 3	**Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour
Corticosteroids	Prednisolone, methylprednisolone	Block cytokine transcription (IL-1, IL-2) and redistribute lymphocytes	Diabetes, bone loss, cataract
Costimulation blocker	**Belatacept**	CTLA4-Ig binds CD80/86 (signal 2)	**PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative
Depleting antibodies	rATG (rabbit antithymocyte globulin), alemtuzumab	T-cell depletion	Infection, PTLD
Non-depleting antibody	**Basiliximab**	Anti-CD25 (IL-2 receptor)	Few

Class	Drug(s)	Mechanism / action	Main toxicity
Antibody / B-cell agents	Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, bortezomib, eculizumab, imlifidase	Antibody removal or blockade; antibody-mediated rejection, desensitisation	—

Pre-transplant assessment of immunological risk

- cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC [complement-dependent cytotoxicity] and flow).

Induction agents

Agent	Mechanism	Use and drawbacks
Basiliximab	Anti-CD25 (IL-2 receptor); non-depleting	Low-risk recipients; 20 mg IV (intravenous) on days 0 and 4; few side effects
rATG	Polyclonal T-cell depletion	High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD
Alemtuzumab	Anti-CD52; profound depletion	Allows steroid avoidance; off-label
IV methylprednisolone	Cytokine suppression	Given intraoperatively with all regimens

- **Over 90%** receive induction, then **tacrolimus + mycophenolate** ± steroids; steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.
- In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

Common protocols

1. **Standard-risk (most recipients):** basiliximab 20 mg IV on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction (see below).
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.
5. **ABO-incompatible or HLA-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive (see below).
6. **Rejection:** T-cell-mediated: pulse IV methylprednisolone, then rATG (for Banff grade II–III); antibody-mediated: steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring: tacrolimus trough levels** (higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).

Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG** after each session; or **high-dose IVIG** (immunomodulation).

- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.
- **Alternatives: kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

Steroid-free immunosuppression

Why avoid steroids

- **Metabolic:** PTDM (post-transplant diabetes mellitus), hypertension, dyslipidaemia, weight gain, hence **cardiovascular death** (leading cause of death with a functioning graft).
- **Bone:** osteoporosis, avascular necrosis; **growth failure** in children.
- Cataract, infection, poor healing.

Strategies

Strategy	Timing
Steroid avoidance	No steroids, or perioperative doses only
Early (rapid) withdrawal	Stopped within the first week; most studied
Late withdrawal	Months later; more rejection, less favoured

Regimen

- **Induction is essential:** depleting rATG or alemtuzumab; basiliximab in low risk.
- **Maintenance:** tacrolimus at target levels plus MMF (mycophenolate mofetil).

Selection

- **Suitable:** low immunological risk: first transplant, non-sensitised, no DSA, prompt graft function, good adherence.
- **Unsuitable:** highly sensitised, DSA, retransplant after immunological loss, DGF, early rejection, **recurring steroid-responsive disease** (IgA nephropathy, lupus), steroids needed for other illness.

Evidence

- **Woodle et al. (2008):** withdrawal after day 7 versus low-dose maintenance: similar 5-year outcomes; **more mild, steroid-sensitive rejection**.
- **HARMONY (Lancet 2016):** steroids stopped on day 8: similar rejection and graft survival, **less PTDM**.
- **Cochrane review (2016):** more acute rejection, but no difference in patient or graft survival up to 5 years.

- **KDIGO (Kidney Disease: Improving Global Outcomes) 2009:** low-risk recipients with induction may stop steroids within the first week; if continued beyond it, continue rather than withdraw.
- **Practice:** over two-thirds still take maintenance steroids (Chapter 91); steroid-free maintenance is increasing in low-risk children.

Risks and safeguards

- Early rejection; possibly more chronic injury and recurrence. Safeguards: tacrolimus levels, adherence, DSA monitoring, biopsy for rising creatinine.

Ciclosporin (cyclosporin A)

Introduction

- Cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*; the first **CNI**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

Mechanism

1. Ciclosporin binds cyclophilin in the cytoplasm.
2. The complex inhibits calcineurin.
3. NFAT (nuclear factor of activated T cells) stays phosphorylated.
4. No IL-2 transcription, so T-cell activation is blocked (signal 1).

Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.
- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy)
Metabolic	Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia
Cosmetic	**Gingival hyperplasia, hirsutism**
Neurological	Tremor, PRES (posterior reversible encephalopathy syndrome)
Immune	Infection, skin cancer, PTLD

Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.

- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

Versus tacrolimus

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.
- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.

Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems with mTOR inhibitors; stones (CNI, steroids): CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).
- **Also know:** belatacept (BENEFIT trial: better GFR), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

Diagrams to draw

- APC (antigen-presenting cell)–T cell synapse with signals 1–3 and the drug acting at each.
- T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.
- Antibody titre falling with plasmapheresis sessions to target before transplant.
- Timeline: induction, steroids stopped by day 7, tacrolimus–MMF maintenance.

> **Key points to remember:**
>
> - Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
> - Basiliximab for low risk; rATG for high immunological risk.
> - CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
> - Belatacept blocks signal 2; avoid if EBV-negative.
> - Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
> - HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
> - ABO-incompatible: rituximab + plasmapheresis to target titre.
> - Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
> - More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
> - Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
> - Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
> - Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

SAQ 4 Evaluation and selection criteria of the living renal donor

P1 · P2 · P3 **merged from 3**

Asked as: P2 SAQ "Pre-operative evaluation of the living (related and unrelated) renal donor" (asked as "Pre operative investigations for renal donor", Summer 2017 Paper II Q3b; "Pre-op assessment of unrelated donor", Practice Set 8 Paper II Q3a) · P3 SAQ "Assessment criteria for kidney donor." (2022-Summer Paper 3 Q3a) · P1 SAQ "Extended Criteria for Selection of Live Renal Donor" (2025-Winter Paper 1 Q3a)

Assumed meaning: evaluation and acceptance criteria for a living kidney donor; for the unrelated question, a living unrelated kidney donor (Indian context); "extended criteria" read as the expanded (marginal, "medically complex") living donor. Related: 2023-Winter Paper 4, Q3e; marginal donor in 2024-Winter Paper 4, Q3e.

Principles

- **Donor safety first. KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold.
- Multidisciplinary team with an **independent donor advocate**.
- For an unrelated donor the medical work-up equals that of a related donor; the extra step is **legal and ethical**: proof of genuine, unpaid motive.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011** (Rules 2014): near relatives may donate; others need authorisation committee approval; swap donation allowed.

Legal assessment of the unrelated donor (THOA)

- A donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

Medical assessment sequence

1. Compatibility (immunological)

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC [complement-dependent cytotoxicity] and flow), DSA (donor-specific antibody). Incompatible pairs: paired exchange.
- Unrelated donor: more HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.

2. Kidney function and urine

- GFR (glomerular filtration rate) by two methods, one measured (e.g. DTPA [diethylenetriamine pentaacetic acid] GFR or creatinine clearance).
- Urinalysis, culture; persistent microscopic haematuria: cystoscopy, imaging, ± kidney biopsy.

3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, coagulation,

BMI (body mass index).

- Active cancer, chronic infection and cardiovascular disease exclude.

4. Infection screen

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

5. Cardiopulmonary: ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

6. Age-appropriate cancer screening: Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

7. Urological: stones (metabolic work-up), haematuria, malignancy, recurrent UTI (urinary tract infection).

8. Anatomy

- **CT (computed tomography) angiography:** number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA [dimercaptosuccinic acid]) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

9. Psychosocial: voluntary, no coercion or payment; independent donor advocate.

Acceptance criteria: standard, extended and exclusion

Factor	Standard (accept)	Extended (individualise)	Exclude (decline)
Age	Adult	Older donors (often >60) acceptable; no upper limit	Under legal age
GFR, mL/min/1.73 m²	≥90	**60–89**	<60
Albuminuria	<30 mg/g ACR (P1: <30 mg/day)	**30–100** mg/g ACR (P1: mg/day)	>100 mg/g ACR (P1: mg/day)
Hypertension	None	Controlled on 1–2 drugs, no target-organ damage	Uncontrolled, organ damage
Obesity	BMI <30	**30–35**; individualised risk	Many centres: >35
Glucose	Normal	Prediabetes; selected type 2 diabetes	Type 1 diabetes
Stones	None	Single past stone, low recurrence risk; small stone removed ex vivo	Bilateral or recurrent; cystinuria, primary hyperoxaluria
Anatomy	Single vessels	Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia)	Major anomaly

- ACR = albumin–creatinine ratio.
- P2 (unrelated-donor answer) states that diabetes, significant obesity and uncontrolled hypertension exclude.

Extended criteria (marginal, medically complex living donor)

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.

- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk. Healthy older donors are acceptable.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype (test in African ancestry); future pregnancy (more pre-eclampsia in later pregnancy).
- **Safeguards:** measured GFR and split function (better kidney stays with the donor); CT angiography; laparoscopic retrieval; THOA authorisation; independent donor advocate.
- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

Counselling figures (Muzaale 2014)

- Function settles at 60–80% of baseline.
- Lifetime ESKD (ESRD, end-stage renal disease) 90 per 10,000 donors vs 14 in matched healthy non-donors; mortality 3 per 10,000.

Diagrams to draw

- Donor work-up flowchart: 1. Legal approval. 2. Compatibility. 3. Medical. 4. Anatomy. 5. Surgery.
- Risk-threshold scale (accept, individualise, decline).

> **Key points to remember:**
> - KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
> - GFR 60–89 and albuminuria 30–100 are individualised.
> - Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
> - ABO and crossmatch first; CT angiography for anatomy.
> - Better kidney stays with the donor.
> - THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
> - Interview video-recorded; no commerce.

SAQ 5 CAPD and CAPD peritonitis

P1 · P2 **merged from 2**

Asked as: P2 SAQ "CAPD (Continuous Ambulatory Peritoneal Dialysis)" (Summer 2018 Paper II Q3e; Practice Set 1 Paper II Q3a) · P1 SAQ "Management of CAPD peritonitis" (Practice Set 21 SAQ Sheet Q3d)

Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** (about 1–2 m^2) as the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water by osmosis: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).

- **CAPD regimen:** about **4 exchanges of 2 L daily**, 4–6-hour dwells, overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff about 2 cm from the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Delay full volumes about 2 weeks to prevent leaks.

Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.
- **PET** (peritoneal equilibration test) identifies high transporters, who do better on short dwells (APD).

Advantages over haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Favoured in children, poor vascular access and unstable heart disease.

Contraindications

- **Absolute:** peritoneal adhesions or fibrosis (scarring), loss of membrane function (membrane failure), uncorrectable hernia, nobody able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

Complications (overview)

- **Peritonitis** (see below).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.
- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** the PD (peritoneal dialysis) catheter is removed after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD patients.

CAPD peritonitis: introduction

- The leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

CAPD peritonitis: diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/μL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

CAPD peritonitis: initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

CAPD peritonitis: empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in the long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).
- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

CAPD peritonitis: organism-specific duration

Organism	Duration
Coagulase-negative *Staphylococcus*, streptococci, culture-negative	2 weeks
S. aureus, enterococcus, Gram-negatives	3 weeks
Pseudomonas	3 weeks, two antipseudomonal drugs
Polymicrobial enteric	Add metronidazole; surgical review (bowel perforation)
Fungal	**Immediate catheter removal** + antifungal

CAPD peritonitis: catheter removal

- **Refractory** (effluent not clear after 5 days of the right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

CAPD peritonitis: definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy.
- **Recurrent:** different organism within 4 weeks.
- **Repeat:** same organism after 4 weeks.

CAPD peritonitis: prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

Diagrams to draw

- Tenckhoff catheter: two cuffs (deep and superficial), tunnel, downward exit site, pelvic tip.
- Peritonitis management flowchart from cloudy bag to catheter removal.

Key points to remember:

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

SAQ 6 THOA/THOTA legislation and hospital authorization committee

P2 · P4 **merged from 2**

Asked as: P2 SAQ "THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules" (Amendments to THOA: 2011; HOTA and its modification) · P4 SAQ "Hospital authorization committee for live related kidney donation."

Indian law; not covered in the chapter notes. Assumed meaning (Practice Set 11): India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.

Legal framework and background: THOA 1994 aims

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as **THOTA (Transplantation of Human Organs and Tissues Act)**; **Rules 2014**.
- THOA 1994 regulates removal, storage and transplantation of organs, **prevents commercial dealing** (banned commercial dealing) and **recognised brain-stem death as legal death**, allowing deceased donation.
- Purpose: **prevent organ trade** while allowing genuine donation.
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

Main provisions of THOA 1994

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

2011 amendment (THOTA)

Area	Change	Purpose
Scope	**Tissues** added; renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**; **tissue banks** registered	Regulates tissue banks
Near relatives	**Grandparents and grandchildren** added to spouse, children, parents and siblings	Wider related donor pool

Area	Change	Purpose
Swap donation	**Paired exchange** (swap donation) between near-relative pairs allowed, with Authorisation Committee approval	Overcomes ABO or crossmatch incompatibility
Retrieval centres	Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres**	
Transplant coordinator	Mandatory in every registered transplant hospital	Organises donation
Mandatory ("required") request	ICU doctor or staff must ask families of a potential brain-dead donor about donation and inform the retrieval centre	More deceased donors
Brain-death board	If a neurologist or neurosurgeon is unavailable, a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve	Smaller hospitals can certify
Eye donation	Trained technicians may remove corneas	
Registry and network	National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation)	Fair allocation
Penalties	**Higher penalties** for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore) (see below)	Deterrence

Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore** (P4: imprisonment up to 10 years and fine up to ₹1 crore).
- Registered medical practitioner: name removed from the register (doctors' registration removed) for **3 years** for a first offence and **permanently** for a later offence.

Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

Living donor categories and approval

Donor	Approval
Near relative: spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren	Verified by the hospital's competent authority
Other than near relative (affection or attachment)	**Authorisation Committee approval mandatory**
Swap (paired) donation between near-relative pairs (2011)	Authorisation Committee
Foreign donor or recipient	Authorisation Committee; embassy certifies relationship

Hospital authorisation committee: constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

Hospital authorisation committee: functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).
7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

Safeguards and restrictions

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- **Minors and mentally ill persons** cannot be living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.

Diagrams to draw

- Deceased donation flowchart: potential donor (brain-dead patient) in ICU → required request → transplant coordinator → brain-stem death board (Board of four) → family consent → retrieval → national registry and allocation (NOTTO allocation).
- Living donor approval flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

> **Key points to remember:**
>
> - THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
> - 2011: tissues added; grandparents and grandchildren became near relatives.
> - Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
> - Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
> - Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
> - Hospital-based committee if >25 transplants a year; no transplant team member.
> - Non-relatives, swap and foreigners need committee approval.
> - Interview video-recorded; appeal within 30 days.

SAQ 7 Marginal kidney donor and Remuzzi score

P2 · P3 · P4 **merged from 3**

Asked as: P4 SAQ "Marginal donor and Remuzzi score for pre-implantation biopsy" (Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.) · P2 SAQ "Marginal kidney donor" · P3 SAQ "Marginal Donor"

"Preemptive" is read as pre-implantation (procurement) biopsy.

Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- **Ojo (2001):** ECD transplantation still gives better survival than staying on dialysis.

Deceased marginal donors

1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002

- Age **60 or over**, or
- Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.
- Graft-loss risk about 1.7 times that of a standard donor (about 70% higher).

2. KDPI (Kidney Donor Profile Index) >85%

- Replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Longevity matching:** lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

3. DCD (donation after circulatory death), Maastricht

Category	Description	Type
I / II	Dead on arrival / failed resuscitation	Uncontrolled
III	Arrest after withdrawal of support	**Controlled (commonest)**
IV	Arrest in a brain-dead donor	Either
V	Euthanasia	Controlled
- Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death).		

4. Others

- Donor AKI (acute kidney injury).
- **HCV-positive (HCV-viraemic)** donors with antiviral treatment.
- **Paediatric donors under 15 kg (en bloc).**
- **Excluded:** metastatic or haematological cancer, melanoma, untreatable viral infection.

Living marginal donors

- Older donors, controlled hypertension (older or hypertensive living donors), obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components (glomeruli, tubules, interstitium, vessels), each 0–3: total 0–12.**

Component	0	1	2	3
Glomerular global sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial and arteriolar narrowing	None	Wall thickness < lumen diameter	Wall = or slightly > lumen	Wall far > lumen, severe narrowing or occlusion

Allocation by score

Total	Grade	Use
0–3	Mild	**Single** kidney transplant
4–6	Moderate	**Dual** kidney transplant
7–12	Severe	**Discard**

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

Optimising use of marginal kidneys

- Short cold ischaemia.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without changing long-term survival; normothermic perfusion tests viability.
- **Old-for-old** matching: **ESP (Eurotransplant Senior Program)**, donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction; delayed CNI (calcineurin inhibitor) if DGF.

Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.

Diagrams to draw

- Remuzzi grid with four compartments scored 0–3.
- Maastricht categories on a timeline of circulatory arrest.

Key points to remember:

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

SAQ 8 Renin-angiotensin system

P1

Type: SAQ (10 marks)
Source: 2015-Summer Paper 1, Q3c

Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, Na^+/K^+ balance and GFR (glomerular filtration rate)**.

Cascade

`Angiotensinogen (liver) ↓ renin (juxtaglomerular cells) — rate-limiting Angiotensin I ↓ ACE (angiotensin-converting enzyme), mainly lung Angiotensin II → AT1 / AT2 receptors ↓ ACE2 Angiotensin (1-7) → Mas receptor (protective)`

Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

Renin stimulated by	Renin inhibited by
Low renal perfusion pressure (baroreceptor)	High perfusion pressure
Low NaCl at macula densa	High NaCl at macula densa
Sympathetic β1 activity	Angiotensin II (negative feedback)
Prostaglandins, hypovolaemia	Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs)

Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal Na^+ retention, K^+ loss); direct proximal Na^+ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin**

ratio >1.5 with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

Diagram to draw: Juxtaglomerular apparatus and cascade.

Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

> **Key points to remember:**
> - Renin is rate-limiting; released by low pressure, low NaCl, β1.
> - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
> - Renin ratio >1.5 lateralises; reninoma excision cures most.

SAQ 9 Drugs associated with Acute Renal Failure.

P1

Type: SAQ (10 marks)
Source: 2016-Summer Paper 1, Q3c

Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

Classification by mechanism

Mechanism	Drugs	How
Haemodynamic (prerenal)	**NSAIDs** (non-steroidal anti-inflammatory drugs)	Block prostaglandin afferent dilatation
	ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)	Remove efferent constriction that holds GFR (glomerular filtration rate)
	Ciclosporin, tacrolimus, contrast; diuretics	Afferent vasoconstriction; volume loss
ATN (acute tubular necrosis)	**Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast	Direct tubular toxicity

Mechanism	Drugs	How
AIN (acute interstitial nephritis)	**β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors**	Cell-mediated hypersensitivity
Crystals	Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis	Intratubular obstruction
Microangiopathy	Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors	Endothelial injury
Rhabdomyolysis	Statins (with fibrates)	Myoglobin toxicity, precipitation

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

Recognition

Type	Urine
Prerenal	FeNa (fractional excretion of sodium) <1%, bland
ATN	FeNa >1%, muddy-brown casts
AIN	Sterile pyuria, white cell casts, ± rash, fever, eosinophilia
- AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable.	

Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.
- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).
- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

Diagram to draw: glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

> **Key points to remember:**
> - NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
> - ATN: aminoglycosides, amphotericin, cisplatin, contrast.
> - AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
> - Crystals: aciclovir, methotrexate, indinavir.

SAQ 10 Berger's Disease.

P1

Type: SAQ (10 marks)
Source: 2020-Summer Paper 1, Q3a

Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).

Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

Oxford MEST-C score (biopsy)

Item	Grades
M: mesangial hypercellularity	M0, M1
E: endocapillary hypercellularity	E0, E1
S: segmental sclerosis	S0, S1
T: tubular atrophy/interstitial fibrosis	T0 ≤25%, T1 26–50%, T2 >50%
C: crescents	C0 none, C1 <25%, C2 ≥25% of glomeruli

Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut

mucosal lymphoid tissue.

- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

Diagram to draw: glomerulus with mesangial IgA deposits.

> **Key points to remember:**
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

SAQ 11 Vascular complications after renal transplantation, including transplant renal artery stenosis

P1

Asked as:

- Vascular Complications after Renal Transplantation in Recipient — *Winter 2025 Paper I Q3d (SAQ)*
- Transplant renal artery stenosis — *Practice Set 20 SAQ Q3d (SAQ)*

Type: SAQ (10 marks)

(vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.
- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

Main complications

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Incidence	0.2–5%	0.1–8.2%	1–23% (commonest)
Timing	First days	First week	Months (usually 3 months–2 years)
Presentation	Sudden anuria	Painful swollen graft, haematuria, oliguria	Refractory hypertension, rising creatinine, flash pulmonary oedema
Doppler	No arterial flow	**Reversed diastolic flow**, clot in vein	High velocity at stenosis; **parvus-tardus** waveform distally

	Arterial thrombosis	Venous thrombosis	TRAS (transplant renal artery stenosis)
Treatment	Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable	Immediate exploration, thrombectomy; nephrectomy if necrotic	**Angioplasty or stenting** (first choice); surgical revision if that fails

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

Transplant renal artery stenosis (TRAS) in detail

Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.
- MRA (magnetic resonance angiography) or CO_2 angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.

- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

Prevention

- Anastomose away from atheroma; check the lie for kinking; avoid clamp injury and kinking; use a Carrel patch where possible.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.
- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

Diagram to draw: Doppler waveforms: normal, parvus-tardus (TRAS; distal to stenosis) and reversed diastolic flow (venous thrombosis).

> **Key points to remember:**
> - TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
> - Refractory hypertension plus rising creatinine: Doppler first.
> - Venous thrombosis: reversed diastolic flow, swollen tender graft.
> - Arterial thrombosis: sudden anuria; explore immediately.

SAQ 12 Management of acute hyperkalaemia

P1

Asked as:

- Management of Acute Hyperkalemia — *Winter 2025 Paper I Q3e (SAQ)*
- Mgt. of acute hyperkalemia. — *Practice Set 6 Paper I Q3e (SAQ)*

Type: SAQ (10 marks)

(also answered in 2025-Winter Paper 1, Q3e)

Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.

- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level (high levels) causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, β_2-agonists via Na^+/K^+-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

Causes

Mechanism	Examples
Pseudohyperkalaemia	Haemolysis, tourniquet, fist clenching, thrombocytosis
Reduced excretion	AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus
Shift out of cells	Acidosis, insulin deficiency, β-blockers, suxamethonium
Cell release	**Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush
Urological	**Jejunal conduit** (reabsorbs urinary potassium)

Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.
- **ECG changes:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → ventricular fibrillation (VF). Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

Emergency treatment (in order)

Step	Drug and dose	Onset and effect
1. Protect the heart	**Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged	1–3 min; lasts 30–60 min; does not lower potassium
2. Shift into cells	**Insulin 10 units regular IV with 50 mL of 50% dextrose**	15–30 min; lowers about 0.6–1 mmol/L; check (monitor) glucose (hypoglycaemia)
	Salbutamol 20 mg nebulised	Additive to insulin
	Sodium bicarbonate	Only with metabolic acidosis
3. Remove	Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis)	Hours
	Haemodialysis	Refractory or oliguric, ESKD (end-stage kidney disease)

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

Diagram to draw: flowchart: ECG change or $K^+ \geq 6.5$ → calcium → shift → remove.

> **Key points to remember:**
> - Calcium stabilises the membrane but does not lower potassium.
> - Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
> - Binders, diuretics and dialysis remove potassium; relieve obstruction.

SAQ 13 Hypokalemia

P1

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 1, Q3f

Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

Causes

Mechanism	Causes
Low intake	Malnutrition
Shift into cells	Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis
GI (gastrointestinal) loss	Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it)
Renal loss, high BP (blood pressure)	**Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess)
Renal loss, normal BP	Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.

- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.
- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.
- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

Diagram to draw: ECG strip showing flat T and U wave.

> **Key points to remember:**
> - Shift or loss; urine potassium separates renal from extrarenal.
> - ECG: flat T, U wave.
> - Diversion acidosis: give potassium with or before bicarbonate.
> - Correct magnesium first.

SAQ 14 Lymphocele and postoperative lymphatic leak (chylous ascites)

P1

Asked as:

- Lymphocele — *Practice Set 8 Paper I Q3a (SAQ)*
- Postoperative lymphatic leak — *Practice Set 14 Paper I Q3b (SAQ)*

Type: SAQ (10 marks)

Postoperative lymphatic leak taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.

Definition and types

- **Lymphocele:** a collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.
- **Chylous ascites:** lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks.

Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9% (symptomatic 0.03–26%); most appear within 6 months, peaking at 6 weeks.

- Also after RPLND (retroperitoneal lymph node dissection; lymphocele ≤1.7%), inguinal dissection, nephrectomy.
- **Chylous ascites:** RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection.

Type	Definition	Setting and frequency
Lymphocele	Lymph collection **without epithelial lining**	**PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7%
Chylous ascites	Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks	RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection

- **Risk factors:** extensive dissection, extended templates, suprahilar dissection near the cisterna chyli (L1–L2), post-chemotherapy surgery, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation (re-transplant).

Clinical features

- **Lymphocele:** most are silent (usually silent). Large ones compress:
- iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
- bladder: frequency, urgency, pelvic pressure;
- ureter or graft: hydronephrosis, rising creatinine (graft dysfunction).
- Infection (may become infected): fever, rigors, sepsis.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention. Air or rim enhancement means infection.
- **Aspirate:** lymph has creatinine equal to (equals) serum (urinoma: much higher), lymphocytes; culture excludes abscess. Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

Management

Lymphocele (treat by symptoms, not size alone)

Situation	Treatment
Asymptomatic, <5 cm	Observe; most resolve
Symptomatic	**Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline)
Recurrent or thick-walled	**Laparoscopic fenestration (marsupialisation)** into a peritoneal window (best)
Infected	Antibiotics + percutaneous drainage; exclude urine leak
Laparoscopy unsafe	Open drainage

```
Asymptomatic, <5 cm → observe Symptomatic → percutaneous drain (± sclerotherapy) Recurrent / thick-walled → laparoscopic peritoneal fenestration (best) Infected → drainage + antibiotics; exclude urine leak
```

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, recurs, risks infection).

- Sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

Chylous ascites (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation. - Continue a low-fat diet for **≥4 weeks** after resolution.

Prevention

- **Clip or ligate visible lymphatics** (clips often preferred; no robust data) (cephalad at the renal hilum in RPLND; perivascular at transplant). Spare lymphatics over the external iliac artery (leg lymph). Avoid unneeded suprahilar dissection.
- **Peritoneal interposition flap** fixed to the bladder (peritoneal flap after robotic PLND) reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).
- Drain after transplantation and major nodal dissections.

Diagram to draw: pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

> **Key points to remember:**
> - Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
> - Lymph creatinine = serum; urinoma creatinine higher.
> - Laparoscopic fenestration best (is best); aspiration alone worst.
> - Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
> - Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

SAQ 15 Hyperfiltration injury in AKI and CKD

P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3c

Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

Mechanism

`Nephron loss → afferent dilatation + efferent constriction → raised glomerular capillary pressure and flow (hyperfiltration) → glomerular hypertrophy; podocytes cannot divide → stretch, detachment → proteinuria → FSGS (focal segmental glomerulosclerosis) → tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)`

In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

Urological settings

Setting	Point
Nephrectomy or donation	Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation
Partial nephrectomy, solitary kidney	>50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains
Congenital solitary kidney, MCDK (multicystic dysplastic kidney)	Monitor BP (blood pressure), proteinuria, yearly ultrasound

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.
- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers glomerular pressure.
- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

Diagram to draw: glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

SAQ 16 UW solution and organ preservation solutions

P2

Asked as:

- UW solution. — *Summer 2017 Paper II Q3c (SAQ)*
- Organ preservation solution — *Practice Set 7 Paper II Q3b (SAQ)*

Type: SAQ (10 marks)

Assumed context: preservation of the kidney for transplantation.

Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

Principles of cold preservation

- Cooling / **Hypothermia** to **4 °C** cuts (slows) metabolism to roughly a tenth, but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) loss (depletion) → **Na^+/K^+-ATPase pump failure** → **cell swelling**; anaerobic glycolysis → **acidosis**; calcium influx. On reperfusion, hypoxanthine is converted by xanthine oxidase to **oxygen free radicals** → ischaemia–reperfusion injury, seen clinically as (clinically this appears as) **DGF (delayed graft function)**.
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors. It therefore needs:
- **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
- **Buffers** (phosphate, histidine) against acidosis.
- **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
- **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
- Electrolytes: "intracellular" (high K^+, low Na^+) or "extracellular" (low K^+, high Na^+) type.

UW composition (per litre)

Component	Amount	Purpose
Lactobionate (K^+ salt)	100 mmol	Impermeant anion; prevents cell swelling; chelates calcium
Raffinose	30 mmol	Impermeant trisaccharide
Hydroxyethyl starch	50 g	Colloid; prevents interstitial oedema
KH_2PO_4	25 mmol	Buffer
$MgSO_4$	5 mmol	Membrane stability
Adenosine	5 mmol	ATP precursor
Glutathione	3 mmol	Antioxidant
Allopurinol	1 mmol	Xanthine oxidase inhibitor; fewer free radicals
Insulin, dexamethasone, antibiotic	Small	Additives

Component	Amount	Purpose
- **Intracellular-type**: K^+ about 125 mmol/L, Na^+ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4.		

UW drawbacks

- **Viscous** (starch): slow flush. **High K^+**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

Comparison of solutions

Feature	UW	HTK (Custodiol)	Euro-Collins
Type	Intracellular (high K^+)	Low K^+, low Na^+	Intracellular
Impermeant	Lactobionate, raffinose	Mannitol	Glucose (enters cells; weaker)
Buffer	Phosphate	**Histidine** (strong)	Phosphate
Colloid	HES (hydroxyethyl starch)	None (low viscosity)	None
Others	Antioxidants, adenosine	Tryptophan, ketoglutarate	—

Solution	Type and key components	Points
UW (University of Wisconsin, Belzer)	Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol	Standard for static storage; viscous; flush out high K^+ before reperfusion
HTK (histidine–tryptophan–ketoglutarate, Custodiol)	Low K^+; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol	Low viscosity, fast cooling; large volume; outcomes similar to UW
Euro-Collins	Intracellular; glucose impermeant, phosphate	First generation; largely abandoned
Celsior	Extracellular; lactobionate, mannitol, glutathione, histidine	Designed for heart; also abdominal organs
IGL-1 (Institut Georges Lopez)	UW-like, but polyethylene glycol and reversed Na^+/K^+	Extracellular; less viscous
- Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's (hypertonic) citrate.		

Practical use and methods

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Keep (aim) cold ischaemia short (ideally <24 h / hours): DGF and thrombosis rise with time (beyond).
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based): pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF vs static storage; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%,

without long-term survival gain (the chapter notes no difference in long-term survival). Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** experimental; allows assessment and repair; limited experience.

Diagram to draw: ischaemia(–reperfusion) cascade with the UW / solution component acting at (that blocks) each step.

> **Key points to remember:**
> - Hypothermia slows, not stops, metabolism.
> - Lactobionate and raffinose (impermeants), HES / starch (colloid).
> - Allopurinol, glutathione (antioxidants); adenosine (ATP).
> - High K^+: flush before reperfusion.
> - HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
> - Machine perfusion reduces DGF.

SAQ 17 Pre transplant nephrectomy

P2

Type: SAQ (10 marks)
Source: 2022-Summer Paper 2, Q3e

Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.

Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

Indications

Group	Indication	Reason
Infection	Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis)	Sepsis under immunosuppression
ADPKD (autosomal dominant polycystic kidney disease)	No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety	Space and symptoms
Malignancy	Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash)	Most RCC (renal cell carcinoma) in recipients arises in native kidneys
Hypertension	Uncontrolled despite multiple drugs	Renin-driven
Heavy proteinuria	Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome)	Hides graft proteinuria, hypercoagulability
Others	Massive polyuria, large stone burden	Graft fluid balance, infection

Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

Also know: pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 cmH_2O.

Diagram to draw: enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

> **Key points to remember:**
> - Default: leave native kidneys.
> - Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
> - Retroperitoneal route for PD; leave a normal ureter.

SAQ 18 Post transplant malignancy

P2

Type: SAQ (10 marks)
Source: 2023-Winter Paper 2, Q3c

Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are

antitumour.

- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

Common cancers

Cancer	Key facts
Non-melanoma skin	Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive
PTLD	Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass
Kaposi sarcoma	HHV-8; regresses with less immunosuppression and switch to sirolimus
RCC (renal cell carcinoma)	4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade
Bladder cancer	About **3-fold** risk; more aggressive
Prostate	Same incidence as general population

Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).
- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

Diagram to draw: pie chart of post-transplant cancers (skin largest).

> **Key points to remember:**
> - Skin cancer commonest; squamous more than basal cell.
> - PTLD: EBV-negative recipient of EBV-positive donor, first year.
> - RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
> - Reduce immunosuppression and switch to mTOR inhibitor.

SAQ 19 Anomalies of renal artery.

P2

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 2, Q3c

Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the

IVC (inferior vena cava).

Terminology

Term	Definition
Multiple renal arteries (preferred)	More than one vessel from the aorta to one kidney
Aberrant	Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal)
Accessory	Two or more branches supplying the same segment

Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

Renal arteriovenous fistula

	Congenital (cirsoid AVM (arteriovenous malformation))	Acquired
Share	<25%	Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma)
Features	Bruit, haematuria, high-output failure	Often resolves
Treatment	Selective embolisation	Observe; embolise if bleeding

Diagram to draw: kidney with main artery, lower-pole accessory artery crossing the PUJ.

> **Key points to remember:**
> - Single artery in 71–85%; accessory arteries usually lower pole.
> - Crossing vessel can cause PUJ obstruction.
> - Treat aneurysm >2 cm or in pregnancy.

SAQ 20 Mycophenolate Mofetil.

P2

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 2, Q3b (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

Pharmacology

Feature	Detail
Dose (kidney)	**1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent
Metabolism	Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation**
Excretion	MPAG in urine; accumulates in renal failure
Interactions	**Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine

Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).

- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

Diagram to draw: purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

> **Key points to remember:**
> - Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
> - 1 g twice daily; ciclosporin lowers MPA levels.
> - Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

SAQ 21 Metabolic abnormalities in CKD

P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3e

Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m²) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

Abnormalities

Abnormality	Mechanism	Management
Sodium and water	Fixed urine concentration; sodium retention → overload, hypertension	Salt restriction; diuretics (chlorthalidone works even in advanced CKD)
Hyperkalaemia	Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone	Potassium binders and diuretics, so RAAS blockers can continue
Metabolic acidosis	Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss	Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle
CKD-MBD (mineral and bone disorder)	Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism**	Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary

Abnormality	Mechanism	Management
Anaemia	Erythropoietin falls, mainly below GFR 30	Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13)
Hyperuricaemia	Reduced excretion	Treat gout; asymptomatic lowering unproven (CKD-FIX)
Lipids	High triglycerides, low HDL (high-density lipoprotein)	Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP)
Glucose	Insulin resistance, but slower insulin clearance → hypoglycaemia	Lower insulin doses

Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).
- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

Diagram to draw: CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

> **Key points to remember:**
> - FGF-23 rises first; secondary then tertiary hyperparathyroidism.
> - Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
> - Hyperkalaemia: binders rather than stopping RAAS blockade.

SAQ 22 Renal transplant in neurogenic bladder

P2

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 2, Q3a

Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm H_2O**.
- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

Pretransplant optimisation (stepwise)

`Antimuscarinic (oxybutynin), mirabegron, botulinum toxin → CIC (clean intermittent catheterisation) ± overnight drainage → bladder cycling if defunctionalised → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation`

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

Diagram to draw: stepwise bladder optimisation ladder.

> **Key points to remember:**
> - Target <40 cm H_2O storage before transplant.
> - Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
> - Pyelonephritis, not augmentation, loses grafts.

SAQ 23 Pre-emptive renal transplant

P2

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 2, Q3a

Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m², 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

Also know: ABO-incompatible or paired exchange programmes widen pre-emptive options.

Diagram to draw: Wolfe curve: relative risk of death after transplant vs dialysis over time.

> **Key points to remember:**
> - Transplant before dialysis = best outcome.
> - Refer at eGFR <30; transplant usually at eGFR <20.
> - Commonest with living donors and children.

SAQ 24 Tacrolimus.

P3

Type: SAQ (10 marks)
Source: 2017-Winter Paper 3, Q3c

Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

Mechanism

`Tacrolimus + FKBP12 (FK506-binding protein 12) → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase) → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)`

- Ciclosporin binds **cyclophilin** instead; same end result.

Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

Toxicity

System	Effect
Kidney	**Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy)
Metabolic	**NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia
Neurological	**Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome)

System	Effect
Others	Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder))

Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

Versus ciclosporin

	Tacrolimus	Ciclosporin
Acute rejection	Lower	Higher
Diabetes, tremor, alopecia	More	Less
Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension	Less	More
- **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens.		
- **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis).		

Diagram to draw: T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

SAQ 25 Urological Complications post Kidney Transplantation.

P3

Type: SAQ (10 marks)
Source: 2023-Winter Paper 3, Q3d

Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

Complications

Complication	Key facts	Management
Urine leak (1.2–8.9%; commonest early)	Distal ureteric necrosis; lost lower-pole artery; technical	Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large
Ureteric stricture (0.5–9%; commonest beyond 3 months)	Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus)	Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term**
Lymphocele (symptomatic 0.03–26%; peak 6 weeks)	Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors	Treat only if symptomatic; **laparoscopic peritoneal fenestration** best
VUR (vesicoureteric reflux)	50–86% radiological; <1% significant	Treat cause; endoscopic injection low grade; reimplant
Haematuria	Anastomotic; persistent after biopsy = AV fistula	Self-limiting; selective embolisation
Stones (about 1%)	Hyperparathyroidism; calcineurin inhibitors	As native kidney; watch silent obstruction
UTI (commonest infection)	Stent, reflux, steroids	Treat 14–21 days if pyelonephritis

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

Diagram to draw: transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

> **Key points to remember:**
> - Ureter is the weak point; ischaemia is the cause.
> - Leak early, stricture late; open repair best.
> - Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

SAQ 26 Female kidney donors: special precautions

P4

Asked as:

- Female kidney donors: enumerate special precautions — *Winter 2023 Paper IV Q3d (SAQ)*
- Female kidney donors: enumerate special precautions — *Practice Set 4 Paper IV Q3a (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3d)

Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.
3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months (another source: often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

Diagram to draw: flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
>
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

SAQ 27 Laparoscopic donor nephrectomy (including pros and cons)

P4

Asked as:

- Laparoscopic donor nephrectomy — *Practice Set 2 Paper IV Q3a (SAQ)*
- Pros and cons of lap donor nephrectomy — *Practice Set 2 Paper IV Q3d (SAQ)*

Type: SAQ (10 marks)
(living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d) **See Practice Set 2 Paper 4, Q3a (full answer).**

Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

Techniques

Technique	Features
Pure transperitoneal laparoscopic	3–4 ports; colon reflected; Pfannenstiel extraction
Hand-assisted (about two-thirds of US cases)	Hand port is also the extraction site; tactile feedback, quick haemostasis
Retroperitoneoscopic	Balloon-made space; no bowel mobilisation; direct hilum; less room
Robotic; LESS (laparoendoscopic single-site surgery)	Wristed dissection; single umbilical incision; cost

Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

Pros and cons (versus open)

Pros	Cons
Less pain, shorter stay, faster (earlier) return to work	Longer operation; learning curve
Better cosmesis; **more people volunteer to donate** (more willing donors)	Slightly longer warm ischaemia
Magnified view of hilum	Shorter vessels, especially right vein (on the right)
Equal graft function and survival (graft function equals open)	Pneumoperitoneum lowers renal blood flow; cost

- **Risks (safety):** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

Diagram to draw: left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

> **Key points to remember (including pros and cons angle):**
> - Ratner 1995; left kidney preferred (longer vein).
> - Hand-assisted in about two-thirds of US cases.
> - Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
> - Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
> - Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
> - Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

SAQ 28 Artificial kidney

P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3e

Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.

Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

Principles

Process	Mechanism	Removes
Diffusion	Concentration gradient across membrane	Small solutes (urea, potassium)
Convection (haemofiltration)	Solute dragged with water	Middle molecules
Ultrafiltration	Hydrostatic pressure gradient	Water
Adsorption	Binding to membrane or sorbent	Some proteins, toxins

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.
- **Transplantation remains the best renal replacement therapy.**

Newer artificial kidneys

Device	Principle	Status
WAK (wearable artificial kidney)	Belt-worn; sorbent regenerates a small volume of dialysate	Early human trials
Wearable or portable peritoneal dialysis	Continuous sorbent-regenerated peritoneal fluid	Development
Implantable bioartificial kidney	Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump	Preclinical
RAD (renal assist device)	Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis	Phase II in acute kidney injury; not adopted
Decellularised or organoid kidneys; pig kidney xenotransplant	Regenerative and xenograft routes	Experimental (see Practice Set 6 Paper 4, Q2)

Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

Diagram to draw: hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

SAQ 29 Robotic kidney transplantation

P4

Type: SAQ (10 marks)
Source: Practice Set 6 Paper 4, Q3f

Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

Diagram to draw: pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

> **Key points to remember:**
> - Vattikuti–Medanta technique with regional hypothermia (2014).
> - Vein first, then artery, then Lich–Gregoir.
> - Best for obese recipients; graft function equals open.

Rapid Revision Sheet · Section 15

Renovascular hypertension: pathophysiology, evaluation and treatment (LAQ)

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

Allograft rejection and immunobiology of renal transplantation (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

Deceased donor kidney: selection, maintenance and retrieval techniques (LAQ)

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

Vascular grafts and vascular surgery in urology (LAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
- Bench surgery plus autotransplantation for complex renovascular lesions.

Vascular anastomosis and vascular access for dialysis (LAQ)

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal: banding or DRIL.

Describe in details basic patho-physiology of urine production and transport (LAQ)

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of Na^+; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL (LAQ)

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) (LAQ)

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) (LAQ)

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.

- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

Acute renal failure (AKI) and acute tubular necrosis (LAQ)

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

Brain-stem death: pathophysiology, guidelines and declaration (SAQ)

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

GFR estimation in prospective renal donor (SAQ)

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens (SAQ)

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.

- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

Evaluation and selection criteria of the living renal donor (SAQ)

- KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
- ABO and crossmatch first; CT angiography for anatomy.
- Better kidney stays with the donor.
- THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
- Interview video-recorded; no commerce.

CAPD and CAPD peritonitis (SAQ)

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: two of cloudy fluid or pain, WCC >100/μL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

THOA/THOTA legislation and hospital authorization committee (SAQ)

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

Marginal kidney donor and Remuzzi score (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

Renin-angiotensin system (SAQ)

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

Drugs associated with Acute Renal Failure. (SAQ)

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

Berger's Disease. (SAQ)

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

Vascular complications after renal transplantation, including transplant renal artery stenosis (SAQ)

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

Management of acute hyperkalaemia (SAQ)

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

Hypokalemia (SAQ)

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

Lymphocele and postoperative lymphatic leak (chylous ascites) (SAQ)

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

Hyperfiltration injury in AKI and CKD (SAQ)

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

UW solution and organ preservation solutions (SAQ)

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High K^+: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low K^+.
- Machine perfusion reduces DGF.

Pre transplant nephrectomy (SAQ)

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

Post transplant malignancy (SAQ)

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

Anomalies of renal artery. (SAQ)

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

Mycophenolate Mofetil. (SAQ)

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

Metabolic abnormalities in CKD (SAQ)

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

Renal transplant in neurogenic bladder (SAQ)

- Target <40 cm H_2O storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

Pre-emptive renal transplant (SAQ)

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

Tacrolimus. (SAQ)

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.

- ELITE-Symphony: low-dose tacrolimus best.

Urological Complications post Kidney Transplantation. (SAQ)

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Female kidney donors: special precautions (SAQ)

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

Laparoscopic donor nephrectomy (including pros and cons) (SAQ)

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

Artificial kidney (SAQ)

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
- Wearable, implantable bioartificial and RAD devices are investigational.

Robotic kidney transplantation (SAQ)

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 16 · BPH

Long answer questions (LAQ)

LAQ 1 Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate

P1 · P3 · P4 **merged from 3**

Asked as: P1 LAQ "Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer" (Anatomy of prostate and its neurovascular bundles with its clinical details; Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate) · P4 LAQ "Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy" (Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate; also SAQ: Neurovascular bundle preservation in Radical prostatectomy: recent techniques) · P3 SAQ "Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate"

"if localised" is read as "of localised" prostate cancer.

Introduction

- A fibromuscular, glandular organ of about 20 g around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Modern RP is anatomical: each anatomical step or new technique either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

Zonal anatomy (McNeal 1981)

Zone	Share of glands	Position	Clinical / surgical relevance
Peripheral (PZ)	About 70%	Posterior and lateral; longest ducts	**70% of cancers**; felt on DRE; biopsy target; posterolateral, so ECE (extracapsular extension) runs along the nerves

Zone	Share of glands	Position	Clinical / surgical relevance
Central (CZ)	About 25%	Cone around the ejaculatory ducts at the base	Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia)
Transition (TZ)	About 5%	Two lobes beside the preprostatic sphincter; shortest ducts	**BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%) missed by DRE
Periurethral glands	Small	Within the preprostatic sphincter	Middle-lobe BPH
Anterior fibromuscular stroma (AFMS)	No glands	Anterior	Anterior tumours reach it; anterior margin risk in posterior approaches

- **Capsule:** there is **no true capsule**, so cancer breaching the gland shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

Fascial investments and dissection planes

Layer	Position	Surgical rule
Prostatic fascia	On the gland; carries DVC tributaries anteriorly	Stays on the prostate in nerve sparing
Levator fascia	Covers levator; fuses with prostatic fascia as the lateral pelvic fascia	Incised to reach the NVB
Denonvilliers fascia	Between prostate and rectum; thickest at base and vesicles	Excised completely for the posterior margin

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral (posterior) ECE is suspected.

Nerve-sparing plane	Lies	Trade-off
Intrafascial	Under the prostatic fascia (between capsule and prostatic fascia)	Most nerves kept; more positive margins
Interfascial	Between prostatic and levator fascia	**Standard nerve-sparing plane** for organ-confined disease
Extrafascial	Outside levator and Denonvilliers fascia	Widest margin; ED (erectile dysfunction) likely

- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow. Preserve them; they may be the main penile supply.

- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- DVC: CO_2 pneumoperitoneum tamponades it, so robotic blood loss is low.
- **Lymph:** obturator, internal and external iliac, common iliac, presacral nodes.

Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves; beside the rectum, centred on the seminal vesicle tip. It gives the **cavernous nerves** (parasympathetic).
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the capsular vessels mark them.
- It runs posterolaterally in the lateral pelvic fascia **between prostatic and levator fascia**, **3–5 mm from the capsule**, outside Denonvilliers fascia.
- At the apex they lie at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where most nerve injuries occur.
- Distribution varies (is diffuse): main posterolateral bundle, posterior "hammock", anterolateral fibres (the basis of high anterior release).
- **Injury mechanisms:** incision, entrapment (clip or suture), **heat** (thermal), **traction**; so dissect athermally without traction.
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

Continence anatomy

Passive	Active
Smooth muscle of bladder neck and urethra; anterior support / suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC)	Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches that enter it at the apex

Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP) steps:** 1. Incise endopelvic fascia; spare the pubourethral ligament. 2. Ligate the DVC over the urethra before division. 3. Preserve accessory pudendal arteries. 4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.
- Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

Concepts and newer concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.

2. **RARP (robot-assisted RP):** 3D, 10× magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function and early oncology versus open; less blood loss (300 versus 1400 mL) and shorter stay. Surgeon matters more than platform.
3. **Risk-graded (risk-stratified) nerve sparing:** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale), using grade, positive cores, mpMRI (multiparametric magnetic resonance imaging) capsular contact and nomograms, side by side.
4. **NeuroSAFE:** intraoperative frozen section of each (the whole) posterolateral surface; if positive, that NVB is resected. It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT tested its effect on erectile function.
5. **Anterior-support preservation:** - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins (anteriorly), so avoid for anterior tumours. - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data. - Retzius-sparing and Hood are aimed at continence; potency equal at 12 months.
6. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery (cold division); control the DVC distally or late. Posterior (Rocco) reconstruction may aid early continence.
7. **PLND (pelvic lymph node dissection) templates:**

Template	Nodes	Drainage captured
Limited	Obturator	Least
Standard	+ External iliac	About 38%
Extended	+ Internal iliac	About 63%
+ Common iliac	+ Common iliac	About 75%

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival (staging, not therapeutic); lymphocele is the commonest complication. 8. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge. 9. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.
- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

Nerve-sparing techniques (NVB preservation): recent techniques

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE, perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".

- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- Graded nerve sparing, NeuroSAFE, Retzius-sparing and Hood: see above.

Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

Diagram to draw

- Axial prostate showing the three fasciae, the NVBs at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

> **Key points to remember:**
> - PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
> - NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
> - Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
> - Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
> - Veil and high release for low-risk sides (Tsuzuki).
> - NeuroSAFE frozen section: more sparing, safe margins.
> - Retzius-sparing: earlier continence, more positive margins.
> - Extended PLND captures about 63% of drainage.

LAQ 2 HoLEP

P1 · P4 **merged from 2**

Asked as: P4 LAQ "HoLEP (holmium laser enucleation of the prostate)" (Holep; also SAQ: HOLEP.) · P1 SAQ "HoLEP in BPH"

Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).

Introduction and principle

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG (holmium:yttrium–aluminium–garnet) = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water** (absorbed by water); **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and coagulates small and medium vessels.
- Flexible quartz fibres; **normal saline** irrigation, so no TUR (transurethral resection) syndrome.

Laser	Wavelength	Chromophore	Penetration
Holmium:YAG	2120 nm	Water	0.4 mm
Thulium fibre	1940 nm	Water	0.2 mm
KTP (potassium-titanyl-phosphate, green light)	532 nm	Haemoglobin	0.8 mm
Nd:YAG	1064 nm	–	About 1 cm (sloughing)

MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA recommendation, Statement 43).
- Large glands otherwise needing open prostatectomy (alternative to open prostatectomy for large glands); retention; failed medical therapy.

Technique (steps)

- **Equipment:** 60–150 W Ho:YAG laser; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** about 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):** 1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision. 2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex; **early apical release** to protect (spare) the sphincter. 3. Capsular plane developed anteriorly to 11–1 o'clock (round to 12 o'clock); **apical mucosal strip incised early** to avoid sphincter traction. 4. Anterior commissure divided; lateral lobes enucleated (detached) into the bladder (**en bloc, 2-lobe or 3-lobe**). 5. **Haemostasis** with a defocused beam.
- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop (less bleeding), shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable at **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** for large glands (Naspro 2006; Kuntz 2008, 5 years): equal function, less morbidity, less blood loss, shorter catheter time and stay.
- **Durable; retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP (enucleation) as a first-line alternative to TURP and to open surgery for large (>80 mL) glands.

Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient (early) SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture; urethral stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

Limitation: learning curve

- The main barrier is the **steep learning curve**: from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

Recent advances

- High-power **Moses pulse modulation**; **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving (ejaculation-sparing) and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

Diagram to draw

- Sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

> **Key points to remember:**
>
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
> - Size-independent; best (preferred) in anticoagulated men.
> - Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
> - Retrograde ejaculation about 75%; learning curve about 50 cases.

LAQ 3 BPH: medical management and surgical modalities

P2

Asked as:

- Describe medical management of BPH and role of various surgical modalities in its management. — *Summer 2015 Paper II Q1 (LAQ)*
- Medical treatment of BPH — *Practice Set 6 Paper II Q3f (SAQ)* (also answered in 2015-Summer Paper 2, Q1)

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction, pathophysiology and rationale

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent / DHT-driven gland bulk) and a **dynamic** component (dynamic part: α1A-mediated smooth-muscle tone; static part: DHT-driven gland bulk). Drugs target each. The bladder responds with detrusor overactivity or later underactivity.

Evaluation (before treatment)

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention (PVR >300 mL = chronic retention).
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

Medical management

Option / class	Doses	Key facts
Watchful waiting	—	Mild or non-bothersome LUTS; lifestyle advice
α1-blockers (first line)	Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate)	First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery (do not prevent AUR or surgery). Hypotension, ejaculatory dysfunction; silodosin causes most anejaculation. **IFIS (intraoperative floppy iris syndrome)**
5-ARIs (5α-reductase inhibitors)	Finasteride 5 mg; dutasteride 0.5 mg	For glands (prostate) >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (another source: ~20%) (maximal at 6 months; act after 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down (↓) 51%. Low libido, ED (erectile dysfunction). Also BPH haematuria
Combination	—	For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events

Option / class	Doses	Key facts
Antimuscarinic or β3-agonist (mirabegron, vibegron)	Add-on	Add for persisting storage symptoms; trials included only men with low PVR (PVR <150 mL)
Tadalafil 5 mg daily	PDE5 (phosphodiesterase-5) inhibitor	LUTS with or without (±) ED; not with nitrates
Desmopressin	—	Nocturnal polyuria (night urine >33% of 24-h output); check (monitor) sodium, especially ≥65 years

Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell (↓) 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination (finasteride-containing arms) reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop (withdrawable) after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

Side effects and important cautions

- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.
- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR**: catheterise (catheter), then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines.

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy (surgery if refractory).

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection), monopolar or bipolar	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe when anticoagulation cannot stop; no histology
HoLEP/ThuLEP (laser enucleation)	Any size	Lowest retreatment, least bleeding

Modality	Size (AUA 2023)	Key feature
Simple prostatectomy (open or robotic)	>80 mL	Complete removal; diverticula or stones
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment
iTind (temporary implant)	25–75 mL, no median lobe	Removed after a few days; modest efficacy
PAE (prostate artery embolisation)	Larger glands (AUA 2023: may be offered)	Local anaesthesia; less effective than TURP; radiation, non-target embolisation

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).
- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI) (another source: 6 months for 5-ARIs), then yearly; surgery if refractory.

Diagram to draw: Size-based surgical algorithm (≤30, 30–80, >80 mL); algorithm: LUTS → size, PSA, storage symptoms → drug choice.

> **Key points to remember:**
> - α-blockers: rapid relief, no prevention.
> - 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
> - MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
> - AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
> - TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Short answer questions (SAQ)

SAQ 1 Millin's prostatectomy

P1 · P2 **merged from 2**

Asked as: P1 SAQ "Millin's prostatectomy" · P2 SAQ "Millin's prostatectomy"

Definition

- **Retropubic transcapsular (extravesical) simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage (obstructive complications), or when lithotomy is impossible.
- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

Work-up

- Exclude cancer: PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

Steps

1. Lower midline or Pfannenstiel; extraperitoneal approach to the **space of Retzius**.
2. Control the dorsal venous complex: optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to (below) the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter, sparing it.
6. Trim the bladder neck lip; advance bladder neck mucosa into the fossa.
7. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
8. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin (retropubic)	Freyer (suprapubic)
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched by finger
Fossa haemostasis	Directly visible	Less visible
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	(Dorsal) venous bleeding	Cystotomy leak

Complications

- Bleeding (transfusion <5%): control by catheter traction, Malament purse-string, O'Conor capsular plication.
- Urine leak, UTI (infection), ureteric injury, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis) and thromboembolism.
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

Versus TURP (transurethral resection of prostate)

- Complete removal, low (lower) retreatment, no TUR syndrome; but incision, longer stay.
- **Also know:** vesicocapsular variant.

Recent advances and modern alternatives

- HoLEP; laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

Diagram to draw

- Sagittal view of transverse capsulotomy and enucleation plane.

> **Key points to remember:**
> - Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
> - Apex divided under vision; 5 and 7 o'clock sutures.
> - Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

SAQ 2 TUR syndrome

P1 · P3 **merged from 2**

Asked as: P1 SAQ "TUR Syndrome" · P3 SAQ "TUR Syndrome."

Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling; about 1–2% in large series.

Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Even routine resection absorbs 800–1000 mL.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, mannitol or water (all hypo-osmolar; plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- Cardiovascular: **hypertension with bradycardia**, tachypnoea, later hypotension, ECG (electrocardiogram) changes.

- Severe: seizures, coma, herniation, death. **Na <120 mEq/L** is dangerous.

Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol-tagged irrigant** (breath test) monitors absorption.

Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

Other settings

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy.

Diagram to draw

- Flowchart: absorption → overload + hyponatraemia → cerebral oedema.

> **Key points to remember:**
> - Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
> - Glycine: ammonia and transient blindness.
> - 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
> - Bipolar saline TURP prevents it.

SAQ 3 TURP instruments, techniques and monopolar vs bipolar

P1 · P2 **merged from 2**

Asked as: P1 SAQ "TURP instruments and techniques" (TURP instruments; TURP techniques) · P2 SAQ "Monopolar v/s Bipolar TURP."

Introduction

- **TURP (transurethral resection of the prostate):** an electrified (wire) loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. The benchmark BPH (benign prostatic hyperplasia) operation; gold standard for

glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

Resectoscope

Part	Description
Outer sheath	24–28 Fr; **insulated beak** prevents arcing to the urethra
Inner sheath	Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying
Obturator	Blunt or visual; atraumatic passage
Working element	Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in
Telescope	**Hopkins rod-lens**, usually **30°** (12° alternative)

Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**. The plasma loop cuts at lower temperature and seals vessels.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP [transurethral incision of the prostate]).

Energy: generator and waveforms

- **Generator:** about 500 kHz, too fast to stimulate nerves.
- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

Monopolar versus bipolar: principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding (dispersive) pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument (the loop), so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.
- **AUA:** no preference between monopolar and bipolar; choose by expertise.

Monopolar versus bipolar: comparison

Feature	Monopolar	Bipolar
Current path	Loop → patient → **dispersive pad**	Active and return poles on the loop; stays within the loop
Irrigant	**Non-conducting, hypo-osmolar:** glycine 1.5%, sorbitol, water	**Normal saline**
TUR (transurethral resection) syndrome	Yes; keep resection <90 min	Practically (virtually) eliminated
Resection time	Limited (speed over haemostasis)	No time pressure; large glands
Loop	Tungsten	Platinum–iridium (Plasma)

Feature	Monopolar	Bipolar
Bleeding, transfusion, clot retention	Higher	Lower
Catheter time, hospital stay	Longer	Shorter
Obturator reflex	Risk	Reduced
Efficacy: IPSS (International Prostate Symptom Score), QoL, PVR	Equivalent (Qmax marginally better with bipolar)	Equivalent
Stricture, BNC (bladder neck contracture), sexual function, retreatment	Similar	Similar

Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption); lowest height giving vision.

Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

Troubleshooting a loop that will not cut

`Scope–generator connection → foot pedal → irrigant matches generator (glycine for monopolar, saline for bipolar) → patient grounding (pad contact)`

TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

Outcomes and evidence

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.
- Meta-analyses: monopolar and bipolar have equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).

Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.
- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

Diagrams to draw

- Assembled resectoscope labelling sheaths, working element, loop and flow channels.
- Resection field with clock positions, bladder neck, verumontanum and sphincter.
- Monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
>
> - Continuous-flow 24–28 Fr sheath, 30° telescope.
> - Tungsten loop (monopolar) vs platinum–iridium (bipolar).
> - Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
> - Bladder neck to verumontanum, depth to capsule.
> - Median lobe first; apex last.
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

SAQ 4 Lasers in BPH, including green light

P2 · P3 · P4 **merged from 3**

Asked as: P3 SAQ "Lasers in BPH, including HoLEP" (Various lasers in treatment of BPH; HOLEP) · P2 SAQ "LASERS in BPH." · P4 SAQ "Green light lasers"

HOLEP read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal). Coagulation lasers are obsolete.

Lasers used in BPH

Laser	Wavelength	Chromophore	Penetration	Procedure
Ho:YAG (holmium:yttrium-aluminium-garnet), pulsed	~2120 nm	Water	0.4 mm	**HoLEP** (holmium laser enucleation of the prostate)
Thulium:YAG (continuous) / **TFL** (thulium fibre laser)	2013 / 1940 nm	Water (TFL at absorption peak, about 4× holmium)	0.2 mm	ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection), vaporisation
KTP (potassium titanyl phosphate) / LBO (lithium triborate)	532 nm	Haemoglobin	0.8 mm	**PVP** (photoselective vaporisation), 120–180 W
Diode	940–1470 nm (various)	Water and haemoglobin (mixed)	0.5–5 mm	Vaporisation, enucleation
Nd:YAG (neodymium)	1064 nm	Weak	About 1 cm	Coagulation; abandoned (sloughing)

Enucleation versus vaporisation

Feature	Enucleation (HoLEP, ThuLEP)	Vaporisation (PVP)
Tissue	Whole transition zone to the capsule, then morcellated	Removed layer by layer
Histology	Yes	No
Gland size	**Any (size-independent)**	Up to 80 cc
Retreatment	Rare	Higher, especially >80 cc

HoLEP

- **HoLEP** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG, 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.
- **Indications: size-independent** (<30, 30–80, >80 mL), with or without median lobe; **anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.

- **Steps:** 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **about 2.0 J × 40–50 Hz** for enucleation. 1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock. 2. Perimontanal incision; **early apical release** protects the sphincter. 3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**). 4. Haemostasis with a defocused beam. 5. **Morcellation** with a full bladder; three-way catheter.
- **Limitation: steep learning curve** (about 50 cases).

Green light (KTP/LBO) lasers: physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP or LBO crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly and vaporisation is bloodless.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

Green light platforms

System	Power	Note
KTP	80 W (later 100 W)	Popularised PVP (photoselective vaporisation of the prostate)
LBO HPS (High Performance System)	120 W	Faster vaporisation
LBO XPS	**180 W**	750 µm MoXy fibre: metal tip, fluid cooling

PVP technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

Indications by guideline (AUA)

- HoLEP or ThuLEP for any size.
- PVP for prostates <30 cc and 30–80 cc (under 80 cc); not >80 cc (high retreatment).
- HoLEP, ThuLEP or PVP for men at **higher bleeding risk** (anticoagulated men, who can continue anticoagulants with PVP).

Evidence and outcomes

- **HoLEP: size-independent**; the only endoscopic procedure **superior to TURP** (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate), durable. Less bleeding, shorter catheter time and stay; longer operating time. Retreatment rare (whole transition zone removed).
- **ThuLEP:** outcomes equal to HoLEP.
- **GOLIATH RCT** (randomised controlled trial; 29 European centres, about 270 men, prostate ≤100 cc; 180 W PVP vs TURP): PVP **non-inferior** for IPSS, Qmax and complications at 2

years; shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal. Retreatment high above 80 cc.

Complications

- Dysuria and storage symptoms (dysuria worst after PVP, the highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum after PVP.
- Transient storage symptoms and early / transient stress incontinence (SUI, stress urinary incontinence).
- Retrograde ejaculation in most (**about 75%**, three-quarters, after HoLEP).
- Bladder neck contracture (BNC) 0–3.2% (P2: 0–3%); stricture; capsular perforation (0.2–1% with PVP); ureteric orifice injury; **morcellator bladder injury**.
- No TUR syndrome (saline); transfusion very rare; erectile function unchanged; PVP leaves **no tissue for histology**.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.
- GreenLight enucleation; office and same-day catheter-free pathways.

Diagrams to draw

- HoLEP: sagittal prostate with enucleation plane along the capsule; adenoma enucleated to the capsule, pushed into the bladder, then morcellated.
- PVP: side-firing fibre at 70° vaporising a lateral lobe.

> **Key points to remember:**
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm.
> - HoLEP is size-independent and the only endoscopic method superior to TURP; suits anticoagulated men.
> - Retrograde ejaculation about 75% after HoLEP.
> - Green light: 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration; 180 W XPS with MoXy fibre.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

SAQ 5 Silodosin

P2 · P4 **merged from 2**

Asked as: P4 SAQ "Silodosin" (Silodosin.) · P2 SAQ "Silodosin."

Definition and pharmacology

- **Uroselective α1-blocker**, the **most α1A-selective** agent (well over 100-fold vs α1B, or by another source several hundredfold over α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension** (few blood-pressure effects). Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle.

Dose and contraindications

Item	Detail
Dose	**8 mg once daily with a meal**; no titration
Renal: CrCl (creatinine clearance) 30–50 mL/min	**4 mg daily**
Contraindicated	CrCl <30; severe liver (hepatic) impairment; strong CYP3A4 inhibitors (e.g. ketoconazole)
Caution	Other α-blockers (avoid), PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery

Indications and efficacy

Indication	Evidence
LUTS due to BPE (benign prostatic enlargement)	Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg** (**Chapple 2011**, European trial), not better; effective in severe LUTS
MET (medical expulsive therapy)	Distal ureteric stones 5–10 mm; α-blockers reduce ureteric spasm; meta-analyses show higher, faster expulsion than tamsulosin; **SUSPEND (2015)** found no overall tamsulosin benefit
CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)	α-blocker-naive men with voiding symptoms; of recent onset

- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers (subgroup data).
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.
- **Also know:** before ureteroscopy (easier access); with propiverine for storage symptoms; nocturia (modest).

Adverse effects

- **Ejaculatory dysfunction / anejaculation (signature effect):** over a quarter of men in registration trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop the drug; men with anejaculation had better responses. Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; no excess hypotension in the elderly or with antihypertensives.
- Falls in older men: falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; class effect; warn the ophthalmologist before cataract surgery (stopping the drug does not remove risk).

Comparison with other α-blockers

Drug	Selectivity	Dose	Hypotension	Ejaculatory dysfunction	Efficacy in LUTS
Silodosin	α1A (highest); α1A ≫ α1D, α1B	8 mg OD	Lowest	**Highest**	Equal
Tamsulosin	α1A, α1D > α1B	0.4 mg OD	Low	Moderate	Equal
Alfuzosin	Clinically uroselective	10 mg OD	Low	Lowest	
Naftopidil	α1D	25–75 mg	Some	Less	
Doxazosin, terazosin	Non-selective; titrate	1–8 mg, 1–10 mg	Highest	Low	

Diagram to draw

- α1A in prostate and vas, α1B in vessels, α1D in bladder; prostate and bladder neck showing α1A receptor sites blocked by silodosin.

> **Key points to remember:**
> - Most α1A-selective α-blocker; 8 mg with a meal; no titration.
> - 4 mg if CrCl 30–50; avoid if <30.
> - Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
> - Anejaculation is the signature effect; least hypotension.

SAQ 6 Methods to estimate prostate size

P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3f (TRUS also answered in 2017-Winter Paper 1, Q3e)

Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm³ ≈ 1 g**.

Methods

Method	Technique	Accuracy and use
DRE (digital rectal examination)	Estimate size, consistency	**Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL
PSA as surrogate	Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s)	AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most
TAUS (transabdominal ultrasound)	Full bladder; transverse width and height, sagittal length	Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)**

Method	Technique	Accuracy and use
TRUS (transrectal ultrasound)	Ellipsoid formula; planimetry	**Method of choice**
MRI (magnetic resonance imaging)	T2-weighted ellipsoid or segmentation (manual or automated)	Most accurate and reproducible
CT (computed tomography)	Axial images	Overestimates; poor capsule definition
Cystoscopy	Urethral length, lobes, median lobe	Rough; guides procedure choice

TRUS formulas

Shape assumed	Formula
Ellipsoid (standard)	**π/6 (0.52) × transverse × AP (anteroposterior) × length**
Sphere	π/6 × transverse³
Prolate spheroid	π/6 × transverse² × AP
Planimetry	Sum of serial transverse areas

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

Diagram to draw: TRUS transverse and sagittal views with three measured diameters.

> **Key points to remember:**
> - Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
> - DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
> - Planimetry most precise; MRI most reproducible.

SAQ 7 Catheter free trial

P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3e

Assumed meaning: trial without catheter (TWOC) after acute urinary retention.

Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

Types of AUR

Spontaneous	Precipitated
BPH (benign prostatic hyperplasia) progression	Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs
TWOC fails more often; about 75% later need surgery	Better TWOC outcome; about a quarter need surgery

Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

Diagram to draw: algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

> **Key points to remember:**
> - TWOC is first choice after AUR; success about 60%.
> - ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
> - Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

SAQ 8 Irrigation fluids for TURP.

P2

Type: SAQ (10 marks)
Source: 2017-Summer Paper 2, Q3a

Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

Fluids

Fluid	Osmolality (approx.)	Points
Sterile water	0	Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP
Glycine 1.5%	~200 mOsm/kg	Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy
Sorbitol 3%	~165	Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis
Mannitol 5%	~275	Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure
Sorbitol 2.7% + mannitol 0.54%	~180	Continental Europe
Normal saline 0.9%	~308	Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome
- Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**.		

TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

Diagram to draw: monopolar vs bipolar current pathways.

> **Key points to remember:**
> - Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
> - Glycine 1.5%: ~200 mOsm/kg; transient blindness.
> - Risk: gland >45 mL, time >90 min; bag height 60 cm.
> - 3% saline; ≤12 mEq/L rise in 24 h.

SAQ 9 Dutasteride.

P2

Type: SAQ (10 marks)
Source: 2018-Summer Paper 2, Q3c

Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

Landmark trials

Trial	Result
CombAT (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5)	Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery
REDUCE (Reduction by Dutasteride of Prostate Cancer Events, 4 years)	Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention**

Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

Diagram to draw: testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

> **Key points to remember:**
> - Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
> - Volume down 25%; AUR and surgery down about half; needs 6 months.
> - Double the PSA; CombAT and REDUCE.

SAQ 10 Measures to decrease complications of TURP

P2

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 2, Q3b (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

Intraoperative measures

Complication	Prevention
TUR syndrome (hypo-osmolar fluid absorbed, dilutional hyponatraemia)	**Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections
Bleeding	Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze
Ureteric orifice injury	Identify orifices first; take a large median lobe down first
Perforation	Respect the capsule; visible fat means perforation
Incontinence	**Stop at the verumontanum**; resect apex last in a bloodless field
Stricture	Calibrate or dilate meatus, lubricate generously, correct sheath size
Bladder neck contracture	Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands

Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

Diagram to draw: resection field from bladder neck to verumontanum with the external sphincter marked.

> **Key points to remember:**
> - Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
> - Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
> - Verumontanum is the distal limit.

SAQ 11 Tamsulosin

P2

Type: SAQ (10 marks)
Source: Practice Set 12 Paper 2, Q3d

Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

Receptor basis

Subtype	Site	Effect of blockade
α1a (about 70% in prostate)	Prostate, bladder neck, prostatic urethra, iris dilator	Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome)
α1b	Blood vessels	Hypotension (tamsulosin has 10-fold less affinity)
α1d	Detrusor, spinal afferents	Storage symptom relief

Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

Comparison

	Tamsulosin	Silodosin	Alfuzosin
Selectivity	α1a/α1d	Most α1a	Clinical only
Ejaculatory effect	Moderate	Highest	Lowest

Diagram to draw: prostate cross-section showing α1a receptors in stroma and bladder neck.

> **Key points to remember:**
> - 0.4 mg once daily; α1a/α1d selective; no titration.
> - IFIS highest with tamsulosin; persists after stopping.
> - MET benefit in distal stones 5–10 mm.

SAQ 12 Prostatic stents

P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3f (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

Classification

Type	Examples	Features
Temporary, first generation	Urospiral, Prostakath (spirals)	Steel or gold-plated coil; not epithelialised
Temporary, second generation	**Memokath** (thermo-expandable nitinol); Allium (covered nitinol)	Expands with warm water, softens with cold water for removal
Temporary polyurethane	Intraurethral catheter, Spanner	Bulbar anchor; sphincter left free
Biodegradable	Polylactic acid stents	Dissolve; trial use
Permanent, epithelialising	**UroLume** (woven mesh)	Urothelium covers the mesh; withdrawn

Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.

- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH** (paclitaxel balloon).

Diagram to draw: sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath) vs permanent (UroLume, withdrawn).
> - Need detrusor contractility; never cross the sphincter.
> - Migration, encrustation, ingrowth; iTind is the modern temporary device.

SAQ 13 Role of UDS before prostate surgery

P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3a

Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

Index	Formula	Interpretation
BOOI (bladder outlet obstruction index)	PdetQmax − 2Qmax	>40 obstructed; 20–40 equivocal; <20 unobstructed
BCI (bladder contractility index)	PdetQmax + 5Qmax	>150 strong; 100–150 normal; <100 weak

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

Diagram to draw: ICS nomogram with obstructed, equivocal and unobstructed zones.

> **Key points to remember:**
> - PFS is the only test that separates BOO from DUA.
> - BOOI >40 obstructed; BCI <100 weak.
> - EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
> - UPSTREAM: routine UDS did not change outcomes.

SAQ 14 IPSS score

P3

Type: SAQ (10 marks)
Source: 2022-Summer Paper 3, Q3e

Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

Items

No.	Symptom	Phase
1	Incomplete emptying	Voiding

No.	Symptom	Phase
2	Frequency (again within 2 hours)	Storage
3	Intermittency	Voiding
4	Urgency	Storage
5	Weak stream	Voiding
6	Straining	Voiding
7	Nocturia	Storage

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

Severity	Score
Mild	0–7
Moderate	8–19
Severe	20–35

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

Diagram to draw: IPSS chart with scoring columns.

Key points to remember:

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

SAQ 15 Laparoscopic Trans vesical Prostatectomy

P3

Type: SAQ (10 marks)
Source: 2025-Winter Paper 3, Q3d

Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).

Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic

incision, pneumovesicum) allows same-day discharge.

Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

Diagram to draw: sagittal view of cystotomy, bladder-neck incision and enucleation plane.

> **Key points to remember:**
> - Indication: gland >80 g, especially with median lobe, stones or diverticulum.
> - Posterior plane first; divide urethra proximal to verumontanum.
> - 5 and 7 o'clock sutures plus mucosal advancement.
> - Same function as open, less morbidity.

SAQ 16 UroLift (prostatic urethral lift)

P4

Asked as:

- Urolift. — *Winter 2024 Paper IV Q3d (SAQ)*
- Urolift. — *Practice Set 6 Paper IV Q3b (earlier version) (SAQ)*

Type: SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3d)

Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).

Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull (retract) the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

Implant and mechanism

- Spring-loaded implant with T-shaped ends:
- **Capsular tab:** nitinol.
- **Urethral end:** stainless steel (resists encrustation).
- **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.

- **Avoid:** obstructing median lobe (MedLift data non-randomised), large gland, active infection.

Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

Evidence

	L.I.F.T. (2013)	**BPH6 (2015)**
Design	PUL vs sham, 2:1, 19 centres	PUL vs TURP (transurethral resection of prostate), 10 European centres
Result	IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction	Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6%
Durability / retreatment	13.6% surgical retreatment at 5 years	13.6% vs 5.7% at 2 years; TURP better IPSS, Qmax, PVR (post-void residual)

- TURP wins on IPSS, Qmax and PVR (post-void residual).

Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction (misfire).
- Lower major adverse events but **higher retreatment than TURP**.

Diagram to draw: axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins (better) on IPSS, flow and durability.

Rapid Revision Sheet · Section 16

Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate (LAQ)

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

HoLEP (LAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
- Size-independent; best (preferred) in anticoagulated men.
- Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

BPH: medical management and surgical modalities (LAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

TUR syndrome (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

TURP instruments, techniques and monopolar vs bipolar (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
- Bladder neck to verumontanum, depth to capsule.

- Median lobe first; apex last.
- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

Lasers in BPH, including green light (SAQ)

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
- Retrograde ejaculation about 75%.
- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

Silodosin (SAQ)

- Most α1A-selective α-blocker; 8 mg with a meal; no titration.
- 4 mg if CrCl 30–50; avoid if <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect; least hypotension.

Methods to estimate prostate size (SAQ)

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Catheter free trial (SAQ)

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

Irrigation fluids for TURP. (SAQ)

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

Dutasteride. (SAQ)

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

Measures to decrease complications of TURP (SAQ)

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

Tamsulosin (SAQ)

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

Prostatic stents (SAQ)

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

Role of UDS before prostate surgery (SAQ)

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

IPSS score (SAQ)

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

Laparoscopic Trans vesical Prostatectomy (SAQ)

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

UroLift (prostatic urethral lift) (SAQ)

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins (better) on IPSS, flow and durability.

Section 17 · Prostate Cancer

Long answer questions (LAQ)

LAQ 1 Staging and management of locally advanced (non-organ-confined) CA prostate

P1 · P2 **merged from 2**

Asked as: P2 LAQ "Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy" (Describe staging of CA prostate and management of non-organ confirmed CA prostate; Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy.; Describe management of LAPC) · P1 LAQ "Describe staging of CA prostate and management of nonorgan confined CA prostate"

Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease; describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy. LAPC is read as locally advanced prostate cancer.

Introduction and definition

- **Locally advanced CaP (carcinoma prostate) / LAPC: cT3–4, N0 or N1 (N+), M0:** extracapsular extension (cT3a), seminal vesicle invasion (cT3b), fixed or invading adjacent organs (cT4), ± pelvic nodes.
- At least 10% of new prostate cancers are locally advanced; these men contribute a disproportionate share of cancer deaths.
- Non-organ-confined disease has a high chance of micrometastases. Any single treatment carries a high recurrence risk (single-modality treatment usually fails), so **multimodal (combined) therapy**, local therapy plus systemic ADT (androgen deprivation therapy), is the rule.
- In the PSA (prostate-specific antigen) era, most "high-risk" disease is defined by grade and PSA rather than palpation.
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort). Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment.

Staging: TNM (tumour, node, metastasis) 2017 (AJCC [American Joint Committee on Cancer] 8th edition)

Stage	Definition
T1a / T1b	Not palpable; incidental at TURP (transurethral resection of prostate): ≤5% / >5% of resected tissue
T1c	Not palpable; found on needle biopsy (for raised PSA)
T2a / T2b / T2c	Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes
T3a	Extraprostatic extension (unilateral or bilateral), including microscopic bladder neck invasion
T3b	**Seminal vesicle invasion** (not T4)
T4	Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall

Stage	Definition
N0 / N1	No nodes / regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites ± bone

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **Prognostic stage groups** combine T, N, M, PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

Risk groups

System	High risk	Very high / locally advanced
D'Amico	T2c, PSA >20 or Gleason 8–10 (any one)	–
NCCN (National Comprehensive Cancer Network)	cT3a, GG4–5 or PSA >20	cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5
EAU (European Association of Urology)	PSA >20 ng/mL, ISUP 4–5 or cT2c	**Locally advanced: cT3–4 or cN+ (cN1), any PSA, any grade**

- Nomograms (CAPRA [Cancer of the Prostate Risk Assessment] score, Partin tables, Kattan nomograms) and genomic classifiers (Decipher) refine risk.

Staging work-up (evaluation)

- **DRE (digital rectal examination)** sets clinical T stage but under- and over-stages. Biopsy ISUP grade group.
- PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 20% if PSA >20 ng/mL and about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** local extent; separates T2 from T3 (extraprostatic extension and seminal vesicle invasion; PI-RADS v2 reporting); best specificity in higher-risk men.
- **Bone scan + CT (computed tomography) or MRI** for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT (positron emission tomography)** is more accurate; **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

Management of locally advanced N0 (cT3–4 N0 M0)

A. Radiotherapy (EBRT, external-beam RT) + long-term ADT: the standard

- Dose-escalated IMRT (intensity-modulated RT) with image guidance, ≥78–81 Gy (P1: about 76–78 Gy), with or without pelvic node RT, with or without brachytherapy boost (**ASCENDE-RT:** LDR [low-dose-rate] boost halved biochemical failure, improving biochemical control; overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and is synergistic with radiation: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk (NCCN; EAU 2–3 years).

Trial	Finding
EORTC 22863	3 years of goserelin with RT improved 5-year OS, 78% vs 62% (other sources: 79% vs 62%)
RTOG 85-31 / 92-02	Long adjuvant ADT better; RTOG 92-02: 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10
TROG 96.01	6 months better than 3 months
SPCG-7; MRC PR07/NCIC PR.3	**RT is still needed with ADT:** adding RT to ADT reduced cancer death and gave an overall survival gain: SPCG-7 cancer death RR (relative risk) 0.44; MRC PR07 OS HR (hazard ratio) 0.77 (P1: HR 0.46). ADT alone is inferior
STAMPEDE (high-risk M0)	Adding 2 years of abiraterone: metastasis-free survival HR 0.53
RTOG 0521; GETUG 12	Docetaxel added modest benefit

- **Duration:** NCCN: ADT 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in high-risk N0.

B. Systemic intensification

- **STAMPEDE (abiraterone, M0):** for N1, or two of cT3–4, Gleason ≥8 or PSA ≥40 ng/mL, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Docetaxel with RT and ADT (RTOG 0521, OS HR 0.69): not routine.

C. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection, ePLND), within multimodal therapy: selected men

- For fit men, life expectancy >10 years, with a completely resectable tumour (mainly cT3a; selected cT3b, limited nodes). Clinical overstaging of cT3 occurs in 7–27%, so cT3 alone is no bar.
- **Advantages:** accurate pathological staging; removes the primary, the source of local morbidity (an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent); RT kept for salvage.
- RP alone gives disease-free survival in at least half at 8–10 years; disease-specific survival 85–92% at 5 years and 79–82% at 10 years (10-year disease-specific survival about 80–90%), but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinical downstaging and fewer positive margins, but **no survival or PSA benefit**.
- **Adjuvant vs early salvage RT after RP:** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 favour adjuvant RT over observation. RADICALS-RT, RAVES and GETUG-AFU 17 favour **observation with early salvage RT** (at PSA rise) for most men with pT3 or a positive margin. Adjuvant RT is still considered for very high-risk pathology (pT3b, GG4–5 / Gleason 8–10, positive margins). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

D. ADT alone

- Only for men unfit for or declining local therapy. Consider immediate ADT if symptomatic, PSA doubling time is short (<12 months) or PSA is very high (>50 ng/mL); otherwise

deferred ADT is acceptable. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.

- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy spares sexual interest (harmful in localised disease); intermittent ADT.

E. Others

- Watchful waiting with deferred ADT in elderly comorbid men.
- Cryotherapy and HIFU (high-intensity focused ultrasound): not standard, investigational in high risk.
- **Also know:** TURP or stents for local obstruction; trial enrolment.

Node-positive disease (N1 M0)

Setting	Management
cN1 M0	RT to prostate and pelvic nodes + long-term ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + ePLND in selected men
pN1 (pN+) after RP	**Immediate ADT** (ECOG 7887: improved overall survival) **or ADT + adjuvant pelvic RT** (beats ADT alone); observation with undetectable PSA and ≤2 nodes

Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

Algorithm

```
Non-organ-confined, cT3–4 / cN1, M0 (PSMA PET or CT + bone scan) ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT ├─ Most (cT3–4 or cN1) → EBRT + ADT 18–36 months │ very high risk / N1 → + abiraterone 2 years └─ Unfit / refuses → ADT alone (immediate if symptomatic, or deferred)
```

Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause bladder outlet obstruction (channel TURP), ureteric obstruction (stent or nephrostomy) and haematuria (palliative RT).

Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

Complications of bilateral orchidectomy

A. Surgical complications

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.

- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

B. Systemic effects of castration

Effect	Details and management
Hot flushes	Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome
Sexual dysfunction	Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men
Osteoporosis, fractures	Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid
Metabolic	Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome
Cardiovascular	Coronary events, sudden cardiac death, thromboembolism
Anaemia	Normocytic; reversible after stopping
Others	Gynaecomastia (≤5%), fatigue, depression, cognitive change

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

Follow-up

- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

Recent advances

- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagrams to draw

- TNM T-stage diagram / sagittal pelvis: T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs.
- Subcapsular orchidectomy.

> **Key points to remember:**
> - Seminal vesicle invasion is T3b; non-regional nodes are M1a.
> - Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
> - RT + long-term ADT (18–36 months; EAU 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
> - EORTC 22863 overall survival 79% vs 62%.
> - STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
> - Neoadjuvant ADT before RP: no survival gain.
> - Early salvage RT preferred over routine adjuvant RT for most men after RP.
> - Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

LAQ 2 Metastatic prostate cancer, including newly diagnosed and bone metastases

P2 · P3 **merged from 2**

Asked as: P2 LAQ "Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid)" (Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain; Management of bone mets in prostatic CA; also SAQ: Zolendronic acid) · P3 SAQ "Management of newly Metastatic prostate cancer"

Zolendronic acid: read as zoledronic acid.

Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).
- **Retention + bone pain case:** this is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases. Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

Definitions

- **mHSPC:** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

Pathophysiology of bone metastases

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; resorbed bone releases growth factors that feed the tumour (**vicious cycle**). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

Evaluation (immediate assessment and work-up)

- **Clinical:** pain, neurological signs, performance status. **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)** define volume; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more and is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending

fracture). **Whole-spine MRI (magnetic resonance imaging)** if any back pain or neurological sign.
- **Histology:** prostate biopsy (or TURP [transurethral resection of the prostate] chips).
- **Germline (and somatic) genetic testing** for all men with metastatic disease (>10% have DNA-repair mutations).
- **Volume (CHAARTED):** see definitions above.

Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent**.
- ADT shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

Step 2: Systemic treatment, hormone-sensitive (mHSPC): start ADT at once, without flare

- **ADT is the backbone, never alone** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist):

Option	Point
Bilateral orchidectomy	Castrate within hours; best for impending cord compression
GnRH (gonadotropin-releasing hormone) antagonist: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix	**No flare**; castrate by day 3 in >95% (degarelix); for impending cord compression
LHRH agonist + antiandrogen	Agonist raises LH (luteinising hormone) up to 10-fold, raising testosterone, for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

Intensified systemic therapy in mHSPC (doublets and triplets)

- ADT alone is no longer standard. Classify volume (CHAARTED) and risk (LATITUDE).

Add to ADT	Trial	Overall survival HR (hazard ratio)
Docetaxel 75 mg/m², 6 cycles	**CHAARTED** (benefit in high volume only); STAMPEDE	0.61; 0.78
Abiraterone 1000 mg + prednisone 5 mg	**LATITUDE**; STAMPEDE arm G	0.62; 0.63
Enzalutamide (ARPI, androgen receptor pathway inhibitor)	ARCHES (radiographic PFS [progression-free survival]); ENZAMET	0.67 (ENZAMET)

Add to ADT	Trial	Overall survival HR (hazard ratio)
Apalutamide (ARPI)	**TITAN**	0.67
Triplet: docetaxel 75 mg/m² × 6 + abiraterone	**PEACE-1** (synchronous)	0.75 (with docetaxel)
Triplet: docetaxel + darolutamide	**ARASENS**	0.68

- Hormonal agents work **regardless of volume**; docetaxel helps only high volume (GETUG-AFU 15 negative), so it is now given within a triplet. Triplets suit chemotherapy-fit men, especially high volume.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

Local and metastasis-directed therapy in mHSPC

- **RT (radiotherapy) to the prostate in low-volume synchronous disease only:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); it also helps local control. HORRAD negative overall.
- Cytoreductive prostatectomy: trials only.
- **MDT (metastasis-directed therapy)** / **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) delays ADT; SBRT also for spinal re-irradiation.

Castration-resistant disease (mCRPC)

- ARPIs, docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 µm, spares marrow): for **symptomatic bone-only** disease (licensed only in castration-resistant disease); 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).
- **^{177}Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

Step 3: Bone pain

- **Analgesia** by the WHO (World Health Organization) analgesic ladder; neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions (30 Gy in 10 fractions, RTOG 9714); retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT.
- **Diffuse pain:** steroids, wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival, licensed only in castration-resistant disease).

Bone-targeted agents

Feature	Zoledronic acid	Denosumab
Mechanism / target	Bisphosphonate: inhibits osteoclast enzyme	Antibody against RANKL
SRE dose (mCRPC) / route	4 mg IV every 4 weeks, adjusted for creatinine clearance	120 mg SC (subcutaneous) every 4 weeks

Feature	Zoledronic acid	Denosumab
Renal adjustment	Yes	No
Evidence	Fewer SREs than placebo	Time to first SRE 20.7 vs 17.1 months vs zoledronate
Toxicity	Fever, myalgia, renal impairment, hypocalcaemia, ONJ	Hypocalcaemia, hypophosphataemia, ONJ 2–4%

- **Neither improves survival**; they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit: CALGB 90202, STAMPEDE). SRE doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Use osteoporosis doses for ADT bone loss: denosumab 60 mg 6-monthly or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

Zoledronic acid (in detail)

- **Introduction:** a potent IV (intravenous) **nitrogen-containing bisphosphonate**.
- **Mechanism:**
- Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
- Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
- Breaks the "vicious cycle".
- Excreted unchanged by the kidneys.

Indication	Dose
SRE prevention, mCRPC (metastatic castration-resistant prostate cancer) and other bone metastases (RCC [renal cell carcinoma], bladder)	**4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance
ADT bone loss (by DEXA and FRAX)	4–5 mg IV yearly
Hypercalcaemia of malignancy	4 mg IV with saline hydration

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC:** no benefit (CALGB 90202, STAMPEDE).
- **Evidence: Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%). **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**
- **Precautions and monitoring:** creatinine before every dose; avoid if creatinine clearance <30 mL/min. **Dental review before starting**; avoid invasive dental work during therapy. Calcium and vitamin D supplements.
- **Adverse effects: acute-phase reaction** (fever, myalgia) after the first infusion; nephrotoxicity, hypocalcaemia, hypophosphataemia; **ONJ**, atypical femoral fracture; ocular inflammation (rare).

Emergencies

Spinal cord compression: watch constantly (every focal back pain is cord compression until proven otherwise): `Back pain ± weakness, sensory level, sphincter change →`

dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly → urgent whole-spine MRI → decompressive surgery + RT (unstable spine, progression on RT, prior RT) or RT alone → orchidectomy or antagonist if hormone-naive (never agonist alone)

- Surgery plus RT beat RT alone for walking ability (**Patchell, 2005**).

Pathological or impending fracture: orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

Algorithms

Retention + bone pain + metastases ├─ Catheter ± nephrostomy; cord signs? → dexamethasone → MRI → RT ± surgery ├─ Orchidectomy / antagonist / agonist + antiandrogen ├─ Analgesia + 8 Gy single-fraction RT; bone protection ├─ Add ARPI or triplet; prostate RT if low volume └─ TWOC after ADT → channel TURP if it fails

- Newly diagnosed mHSPC: split by volume and fitness to doublet or triplet ± prostate RT.

Follow-up

- PSA, ALP, liver tests, calcium, renal function and blood count every 3–6 months; testosterone if PSA rises; new scan if symptoms or PSA rise. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- Manage ADT effects: bone loss (DEXA), diabetes, cardiovascular risk, hot flushes.
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

Diagrams to draw

- The retention + bone pain flowchart above.
- Vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate.
- Algorithm splitting newly metastatic disease by volume and fitness to doublet or triplet ± prostate RT.

Key points to remember:

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief for focal pain; focal back pain = cord compression until MRI proves otherwise; dexamethasone and MRI.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance; no survival benefit.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

LAQ 3 Radiotherapy and brachytherapy for CA prostate

P2 · P4 **merged from 2**

Asked as: P4 LAQ "Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?" · P2 SAQ "Radiotherapy and brachytherapy for CA prostate" (Radiotherapy for CA prostate.; Brachytherapy for CA prostate)

"Focal" is taken to mean localised (organ-confined and locally advanced) disease.

Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**. The low α/β ratio also favours high dose per fraction (HDR brachytherapy).

EBRT (external beam radiotherapy) techniques

Step	Advance
3D-CRT (three-dimensional conformal RT)	CT (computed tomography)-planned beams shaped to the prostate
IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)	Inverse planning, steep dose gradients; less rectal toxicity; now standard; allows dose escalation
IGRT (image-guided RT)	Daily fiducials or cone-beam CT; margins 3–7 mm
MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography)	Better targeting; PSMA PET changes plans in up to half of high-risk men

EBRT fractionation schedules

Schedule	Dose per fraction	Example	Evidence
Conventional	1.8–2 Gy	78 Gy in 39	Dose escalation (RTOG 0126): better PSA control, no survival gain
Moderate hypofractionation (standard)	2.25–4 Gy	**60 Gy in 20**	**CHHiP** (non-inferior to 74 Gy in 37), **PROFIT**: non-inferior
SBRT (stereotactic body RT)	≥5 Gy	36.25 Gy in 5	**HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity

Brachytherapy: definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT can safely give.

Brachytherapy: types

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Source	^{125}I (half-life 59.6 days), ^{103}Pd (17 days), ^{131}Cs (9.7 days)	**^{192}Ir** via remote afterloader
Monotherapy dose	^{125}I 145 Gy (P2: 140–160 Gy); ^{103}Pd 110–125 Gy	27 Gy in 2, 31.5 Gy in 3, 34–38 Gy in 4 fractions (single 19 Gy fraction fails)

Feature	LDR (low dose rate), permanent seeds	HDR (high dose rate), temporary
Monotherapy use	Low and favourable intermediate risk	
Dosimetry / quality	Post-implant CT at 14–30 days: **D90** (dose covering 90% of gland) **>90%** of prescription	Real-time planning; none afterwards

Brachytherapy: technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.
- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 cm^3) limits. Low D90 predicts failure.

Brachytherapy: indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT to implant gave no benefit in intermediate risk, more toxicity).
- **Boost** (EBRT 45 Gy + implant ± ADT [androgen deprivation therapy]) for unfavourable intermediate and high risk: **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, with more grade 3 urinary toxicity (18.4% vs 5.2%).
- Also salvage and focal BT.

Brachytherapy: contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP (transurethral resection of the prostate) defect; gland >60 cm^3 (cytoreduce with ADT); pubic arch interference; prior pelvic RT; inflammatory bowel disease; cannot lie in lithotomy.

Brachytherapy: complications and follow-up

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).
- Nadir is deeper and later (about 2.5 years). **Benign PSA bounce** in about one-third: watch for 30 months before biopsy, do not salvage.

ADT with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

Risk group	ADT with EBRT
Low, favourable intermediate	None
Unfavourable intermediate	**4–6 months**
High / locally advanced	**18–36 months** (P2: 24–36 months) (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE)

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical control), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

Advanced and metastatic disease

Setting	RT role	Evidence
Low-volume de novo mHSPC (metastatic hormone-sensitive)	**RT to prostate** + ADT	**STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume
Oligometastases (≤3–5)	SBRT to all lesions (MDT, metastasis-directed therapy)	STOMP, **ORIOLE** (less progression), SABR-COMET
Painful bone metastases	Palliative EBRT, e.g. **8 Gy single fraction**; cord compression	Equal pain relief to multiple fractions
mCRPC (metastatic castration-resistant), bone-only symptomatic	**Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6	**ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events
mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor)	**^{177}Lu-PSMA-617** (β-emitter radioligand)	**VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED.

Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL.
- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

Toxicity of RT

Type	Features and management
Acute (weeks)	Frequency, dysuria, diarrhoea; α-blockers
Late GI (gastrointestinal)	**Radiation proctitis** with bleeding at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen

Type	Features and management
Late GU (genitourinary)	Cystitis, haematuria, stricture
Sexual	ED, vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most
Rare	Second cancers, fistula

Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; protons; PSMA radioligands moving earlier (PSMAfore).

Diagram to draw

- TRUS-guided perineal template brachytherapy (transperineal implant set-up).

> **Key points to remember:**
>
> - α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
> - ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
> - Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
> - STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
> - Phoenix: nadir + 2 ng/mL.
> - LDR ^{125}I 145 Gy; HDR ^{192}Ir; D90 >90%, V100 >85%.
> - IPSS >20 or Qmax <10 = relative contraindication.
> - ASCENDE-RT: boost halves biochemical failure.

LAQ 4 Castration-resistant prostate cancer (CRPC): definition and treatment

P2

Asked as:

- CRPC in Ca prostate. — *Summer 2020 Paper II Q3a (SAQ)*
- Discuss treatment of CRPC — *Practice Set 6 Paper II Q1 (LAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3a)

Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
- **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
- **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease (visceral disease).

Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue (germline and tumour testing) for HRR (homologous recombination repair) genes such as BRCA2, and MSI (microsatellite instability).
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).
- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

Clinical states model (PCWG3): `Localised → rising PSA → mCSPC (castration-sensitive) ↓ (on ADT) nmCRPC → mCRPC 1st line → 2nd line → 3rd line`

General principles of management

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy; try antiandrogen withdrawal.

2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Sequencing: switch mechanism**; a second ARPI rarely works (cross-resistance). Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival (improve metastasis-free and overall survival):
- Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

mCRPC: life-prolonging options

Class	Agent and dose	Key trial (overall survival)
ARPI (AR-pathway inhibitor)	**Abiraterone** 1000 mg fasting + prednisone	COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy)
	Enzalutamide 160 mg	AFFIRM: 18.4 vs 13.6 months; PREVAIL
Taxane	**Docetaxel** 75 mg/m² 3-weekly + prednisone	TAX 327: 18.9 vs 16.4 months (mitoxantrone)
	Cabazitaxel 20–25 mg/m² after docetaxel	TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI
Radioligand (mCRPC)	**Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral)	ALSYMPCA: 14.9 vs 11.3 months
	^{177}Lu-PSMA-617 (PSMA-positive, after ARPI + taxane)	VISION: 15.3 vs 11.3 months
PARP (poly-ADP-ribose polymerase) inhibitor	Olaparib (HRR genes); rucaparib (BRCA)	PROfound; TRITON3
	Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line)	PROpel; MAGNITUDE; TALAPRO-2
Immunotherapy	**Sipuleucel-T** (asymptomatic, no visceral disease)	IMPACT: 25.8 vs 21.7 months
	Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5%	Tumour-agnostic approval

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

Sequencing by prior mCSPC therapy

Prior therapy	Next option
Docetaxel only	ARPI
ARPI only	Docetaxel, radium-223, sipuleucel-T
Both	Cabazitaxel, ^{177}Lu-PSMA, PARP inhibitor if HRR mutation

Neuroendocrine/aggressive variant

- Neuroendocrine variant: low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks against skeletal events, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.
- Focal bone pain: external-beam radiotherapy (radiotherapy for bone pain).
- **Cord compression** needs high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

Diagram to draw: clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

Diagram to draw: clinical states model with drugs placed at each state.

> **Key points to remember:**
>
> - Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
> - AR still drives CRPC.
> - nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
> - mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
> - Switch mechanism; never radium-223 with abiraterone.

LAQ 5 PSA screening for prostate cancer: evidence, problems and solutions

P2

Asked as:

- What are the various problems in PSA screening — *Practice Set 8 Paper II Q1 (LAQ)*
- Screening of CA prostate. — *Practice Set 13 Paper II Q3d (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

Introduction (concept)

- Screening tests healthy, symptom-free (asymptomatic) men; its aim is lower prostate cancer mortality through earlier treatment (cure). The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, (US) mortality has fallen by >40% (over 40%; SEER data), but at the price of overdiagnosis.

Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).
- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

Evidence from screening trials

Trial	Design	Result
ERSPC (European Randomised Study of Screening for Prostate Cancer)	>182,000 men; PSA every 2–4 years vs none	**20% relative fall** in (prostate) cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented
PLCO (Prostate, Lung, Colorectal and Ovarian), USA	Annual PSA vs usual care	**No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power
Göteborg-1 (Sweden)	Biennial PSA, men 50–64	44% fall in cancer death at 14 years (larger mortality reduction with biennial screening)
CAP (Cluster randomised trial of PSA testing for prostate cancer), UK	Single PSA invitation	No mortality difference at 10 years; only a small fall at 15 years
Göteborg-2 (2022)	MRI-targeted biopsy only	Halved overdiagnosis
- Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years.		

Problems in PSA screening

1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

Guideline positions (guidelines)

Body	Recommendation
USPSTF (US Preventive Services Task Force) 2012	Grade D: against screening

Body	Recommendation
USPSTF 2018 (current)	**Grade C, 55–69 years: individual shared decision; Grade D ≥70** (not ≥70; was Grade D in 2012)
AUA (American Urological Association)	Shared decision 55–69; every 2 years (2-yearly interval)
NCCN (National Comprehensive Cancer Network)	Baseline PSA and discussion at 45
EAU (European Association of Urology)	Risk-adapted early detection: informed men from 50; earlier start (45) with family history or African descent, 40 for BRCA2 carriers
- **No body recommends (advises) unselected population screening.**	
- **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier.	
- Stop when life expectancy is under 10–15 years.	

Solutions: risk-adapted, MRI-first pathway (reducing harm)

`Informed man, life expectancy >10–15 y → PSA (repeat if raised) → risk stratify: PSA density, free/total PSA, risk calculator, PHI / 4Kscore / Stockholm3 → MRI (magnetic resonance imaging) PI-RADS 1–2, low density → no biopsy, keep in follow-up PI-RADS ≥3 → targeted ± systematic biopsy (transperineal) → Grade Group 1 → active surveillance`

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals (intervals by baseline PSA).
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease (Grade Group 1) uncouples overdiagnosis from overtreatment; MRI and biomarkers before biopsy.
- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

Diagram to draw: iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart (pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment).

> **Key points to remember:**
> - ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
> - Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
> - USPSTF 2018: Grade C at 55–69, Grade D ≥70.
> - Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

LAQ 6 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

P2

Type: LAQ (25 marks)
Source: Practice Set 12 Paper 2, Q1 (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.

Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

Definitions of BCR after RP

Body	Definition
AUA (American Urological Association)	PSA ≥0.2 ng/mL
NCCN (National Comprehensive Cancer Network)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL
EAU (European Association of Urology) risk groups	**Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5

Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.
- CT and bone scan are rarely positive below PSA 10 ng/mL.

Treatment

Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI ├ Negative or fossa only → salvage RT to bed + ADT ├ Pelvic nodes only → bed + pelvic node RT + ADT ├ Oligometastatic (≤3) → ADT ± metastasis-directed therapy └ Polymetastatic → treat as mHSPC: ADT + ARPI ± docetaxel **1. Salvage RT (radiotherapy): the only curative option**

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

2. Add ADT (androgen deprivation therapy) to salvage RT: essential here

Trial	Arms	Result
RTOG 9601 (Radiation Therapy Oncology Group)	RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0	12-year OS (overall survival) 76.3% vs 71.3%
GETUG-AFU 16	RT ± 6 months goserelin; PSA 0.2–2.0	10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged
SPPORT (RTOG 0534)	Bed / bed + 4–6 months ADT / pelvis + ADT	5-year freedom from progression 70.9% / 81.3% / 87.4%
RADICALS-HD	None vs 6 vs 24 months ADT	24 months better MFS than 6 months
- At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**.		

3. Systemic therapy

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**
- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.

- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

Diagram to draw: the treatment flowchart above, with PSA kinetics curve after RP.

> **Key points to remember:**
> - EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
> - PSMA PET first: almost 100% detection above PSA 2 ng/mL.
> - Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
> - Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

LAQ 7 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

P2

Type: LAQ (25 marks)
Source: Practice Set 14 Paper 2, Q1 (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).

Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

Investigations

Test	Purpose
Urinalysis, culture	Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms)
Repeat PSA, free PSA	Confirm the rise; low free PSA favours cancer
Creatinine, ultrasound KUB (kidney, ureter, bladder)	Upper tracts, prostate volume (PSA density)
Uroflowmetry, PVR (post-void residual)	Baseline LUTS assessment
mpMRI (multiparametric MRI) before biopsy	PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging
Biopsy	Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis)

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).
- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.
- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

Management of the cancer (life expectancy >25 years)

Situation	Management
Low or favourable intermediate risk	AS (active surveillance) is possible, but uncommon at PSA 18
Localised intermediate or high risk	**RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk
Locally advanced (T3–T4, N1)	Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE)
Metastatic	ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.
- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

```
PSA 18 + LUTS → history, DRE, urine culture → repeat PSA → mpMRI → targeted + systematic biopsy (transperineal) Cancer → risk group + staging → RP / RT + ADT / systemic Benign → explain PSA, re-biopsy if needed → BPH pathway LUTS throughout: α-blocker; no 5ARI until cancer excluded
```

Diagram to draw: the flowchart above.

> **Key points to remember:**
>
> - PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
> - PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
> - Young fit man: RP treats both cancer and obstruction.
> - α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

LAQ 8 Discuss management of organ confined CA prostate

P2

Type: LAQ (25 marks)
Source: Practice Set 15 Paper 2, Q2

Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

Risk stratification (NCCN, National Comprehensive Cancer Network)

Risk group	Criteria
Very low	All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15
Low	All of: T1–T2a, GG1, PSA <10
Favourable intermediate	1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	2–3 intermediate factors, GG3, or ≥50% cores positive
High	Any of: T3a, GG4–5, PSA >20

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

Options by risk

Risk	Preferred options
Very low / low	**Active surveillance (AS)**; watchful waiting if life expectancy short
Favourable intermediate	AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone
Unfavourable intermediate	RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy)
High (organ-confined, e.g. GG4–5 or PSA >20)	RP + extended PLND (multimodal); EBRT + **18–36 months ADT**

1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.
- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).
- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

Landmark trials

Trial	Comparison	Result
SPCG-4 (pre-PSA era)	RP vs watchful waiting	Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years
PIVOT	RP vs observation (PSA era)	No mortality difference; benefit only in intermediate risk
ProtecT	Monitoring vs RP vs RT + ADT	15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

Algorithm

```
Organ-confined CaP → risk group + life expectancy ├─ short life expectancy → watchful waiting ├─ very low / low → active surveillance ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy └─ unfavourable intermediate / high → RP + PLND or EBRT + ADT (4–6 or 18–36 months)
```

Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

Diagram to draw: stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

LAQ 9 Discuss advances in management of CA prostate

P4

Type: LAQ (25 marks)
Source: Practice Set 5 Paper 4, Q2 (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
- **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
- **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
- So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.
- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-guided nerve sparing (NeuroSAFE).
- **Radiotherapy:**
- Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
- **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
- **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
- High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.

- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.
- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

> **Key points to remember:**
> - PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
> - proPSMA: PSMA PET accuracy 92% vs 65%.
> - ProtecT: no mortality difference at 15 years; AS is standard for GG1.
> - VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Short answer questions (SAQ)

SAQ 1 TRUS and prostate biopsy

P1 · P2 **merged from 3**

Asked as: P2 SAQ "Transrectal ultrasonography (TRUS)" (Transrectal ultrasonography; TRUS) · P2 SAQ "TRUS / prostate biopsy" (TRUS biopsy.; Prostate biopsy) · P1 SAQ "Transrectal ultrasound study of prostate"

Read as: transrectal ultrasonography (TRUS) of the prostate, and TRUS-guided prostate biopsy.

Physics, principle and technique of TRUS

- Piezoelectric crystals; higher frequency gives better resolution but less penetration (depth).
- The prostate lies directly in front of the rectum. Probes: **6–10 MHz** end-fire, side-fire or biplane.
- Left lateral position (or lithotomy for transperineal work), after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Transition	5–10% (BPH [benign prostatic hyperplasia])	Heterogeneous, anterior	~20–25%
Central	~25%	Homogeneous, echogenic, at base	~5–10%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone).
- **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles about 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (PSA [prostate-specific antigen] density)** = PSA/volume; **>0.15 ng/mL/cc** is suspicious (suggests cancer).

Findings

- **Cancer:** classically a hypoechoic peripheral-zone lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular (extraprostatic) extension.
- **Cysts:** Müllerian (midline); ejaculatory duct cyst (just off midline); seminal vesicle cyst (lateral; renal agenesis).
- Abscess, calculi.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes diffusely hypoechoic.

Indications for TRUS

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy or cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst).
- Drainage of prostatic abscess or cysts; haematospermia.

Prostate biopsy: introduction

- TRUS-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

Prostate biopsy: indications and contraindications

- Raised PSA with risk factors; suspicious DRE; MRI PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance. Saturation biopsy only after negative biopsies.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

MRI-first pathway

- **PROMIS (2017):** mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.
- **PRECISION (2018):** MRI-targeted biopsy found more significant cancer (38% vs 26%) and avoided 28% of biopsies.
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

Prostate biopsy: preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augmented or swab-directed if risk factors (antibiotics in last 6 months, diabetes, previous infection). Povidone-iodine rectal cleansing.
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the junction of prostate base and seminal vesicle.

Prostate biopsy: technique

	Transrectal	Transperineal
Position	Left lateral	Lithotomy
Needle route	Rectal wall (contaminated)	Sterile perineal skin
Antibiotics	Mandatory	Skin-flora only or none
Anterior/apex	Poor	Good
Anaesthesia	Periprostatic block	Perineal and periapical triangle block

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density).
- 18-gauge spring-loaded needle; indent the capsule, withdraw slightly, then fire, so the core samples peripheral zone.
- **12 cores:** base, mid-gland and apex on each side, medial and lateral (sextant alone is inadequate; it missed lateral cancers). Add MRI-targeted cores (cognitive or fusion). TZ (transition zone) cores if gland >50 mL.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

Complications (transrectal)

Complication	Rate
Haematuria	23–63%
Haematospermia	10–50%
Rectal bleeding	2–22%
Infection	0.1–7%; sepsis 0.3–3%
Retention	0.2–0.4%
Vasovagal reaction	1–5%

- Bleeding is commonest; sepsis is most serious.
- **Sepsis** from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return; prevent with povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.
- Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

Recent advances

- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU 2023.
- **MRI-first pathway** (PROMIS, PRECISION, above).
- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.

- **Micro-ultrasound (29 MHz; 70 µm resolution)** with PRI-MUS scoring, detection comparable to mpMRI.
- **Also know:** PSMA (prostate-specific membrane antigen) PET.

Diagrams to draw

- 12-core biopsy template on a transverse prostate outline.
- Transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

> **Key points to remember:**
> - Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
> - 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
> - Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
> - MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
> - 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
> - Bleeding commonest; sepsis most serious.
> - Transperineal now preferred.

SAQ 2 Multiparametric MRI, PI-RADS and fusion biopsy

P1 · P4 **merged from 2**

Asked as: P1 SAQ "Prostate multiparametric MRI and endorectal coil MRI" (Multiparametric MRI : its components and interpretation for prostate cancer.; Endorectal coil MRI) · P4 SAQ "PI-RADS and MRI–ultrasound fusion biopsy"

Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.
- PI-RADS scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm³ or EPE [extraprostatic extension]).

Components

Sequence	Principle	Cancer appearance
T2W (T2-weighted), triplanar	Anatomy; normal PZ (peripheral zone) bright	Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE and SVI (seminal vesicle invasion)
DWI (diffusion-weighted imaging) with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400)	Cellularity: cellular tumour restricts water movement	Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises
DCE (dynamic contrast-enhanced) T1 with gadolinium	Vascularity (angiogenesis): early enhancement and washout	Early focal enhancement (type 3 curve)

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** image quality score (1–5) before reporting.

Interpretation: PI-RADS v2.1

Score	Clinically significant cancer
1	Highly unlikely
2	Unlikely
3	Equivocal
4	Likely; lesion <1.5 cm
5	Highly likely; ≥1.5 cm or definite EPE

Zone	Dominant sequence	Tie-breaker at score 3
Peripheral	**DWI**	DWI 3 + DCE positive → 4
Transition	**T2W**	T2W 3 + DWI 5 → 4; (v2.1) T2W 2 + DWI ≥4 → 3

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

Clinical use and evidence for MRI

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%); 28% avoided biopsy.
- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent.

MRI–ultrasound fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

	Cognitive	Software fusion	In-bore MRI
Method	Aims from memory of MRI	Software overlay	Needle under MRI
Drawback	Operator-dependent; small lesion in large gland	Cost; registration error	Time, cost

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

Fusion biopsy: indications and evidence

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.
- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

Endorectal coil MRI: principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

Endorectal coil MRI: technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI: **T2W** triplanar for anatomy, **DWI with ADC** map, **DCE**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

Endorectal coil MRI: uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA.
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

ERC versus no ERC

	With ERC	Without ERC (3 T, pelvic coil)
Signal and resolution	Higher	Adequate at 3 T
csPCa (clinically significant cancer) detection	GG (Grade Group) ≥3 equal; GG2 slightly better	GG ≥3 equal
Comfort, cost, time	Worse	Better
Gland	Deformed (fusion and radiotherapy planning harder)	Natural shape

ERC: disadvantages and contraindications

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

ERC: current position

- At **3 T the ERC is largely historical**; PI-RADS v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

Diagrams to draw

- Prostate zones with a PZ lesion dark on ADC.
- MRI target on live TRUS with needle path.
- Sagittal pelvis with inflated endorectal coil against the posterior prostate.

Key points to remember:

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation; cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

SAQ 3 Maximum / combined androgen blockade

P2 · P4 **merged from 2**

Asked as: P2 SAQ "Maximum (combined) androgen blockade and bicalutamide" (Maximum androgen blockade; Bicalutamide) · P4 SAQ "Combined Androgen Blockade."

Definition and rationale

- **MAB (maximum androgen blockade)**, also **CAB (combined androgen blockade)** or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. The antiandrogen blocks these at the AR (androgen receptor).

Antiandrogens

Drug	Dose	Signature toxicity
Flutamide	250 mg three times daily	Diarrhoea, hepatotoxicity
Nilutamide	300 mg, then 150 mg daily	Delayed dark adaptation, pneumonitis
Bicalutamide	**50 mg daily** (with LHRH agent)	Best tolerated; gynaecomastia
Cyproterone (steroidal)	100 mg two to three times daily	Cardiovascular, hepatic; no CAB benefit, not used for CAB now

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

Evidence for MAB

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%** (borderline significance); cyproterone gave no benefit (slightly worse).
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

Current uses of MAB

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold, and testosterone, for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months (mutant AR turns the drug into an agonist).

Bicalutamide: introduction and mechanism

- Non-steroidal **AR antagonist**, called first- or second-generation. It competes with testosterone and DHT at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

Uses of bicalutamide

Use	Dose	Evidence
CAB with LHRH agonist, metastatic	50 mg daily	Small survival gain (2–3%, PCTCG)
Flare prevention	50 mg, with or before the agonist, 2–4 weeks	Main use today
Monotherapy, locally advanced	150 mg daily	**EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23)
With salvage radiotherapy	150 mg for 2 years	**RTOG 9601:** 12-year overall survival 76.3% vs 71.3%
CRPC (castration-resistant prostate cancer)	50 mg	Second-line step; largely replaced by newer agents

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

Adverse effects of bicalutamide

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.

Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

Trial	Addition to ADT (androgen deprivation therapy)	Result
LATITUDE, STAMPEDE (arm G)	Abiraterone + prednisone	OS (overall survival) HR 0.62 / 0.63
ENZAMET	Enzalutamide (vs ADT + older antiandrogen, a CAB control arm)	OS HR 0.67; enzalutamide beat CAB
ARCHES	Enzalutamide	Radiographic PFS (progression-free survival) HR 0.39
TITAN	Apalutamide	OS HR 0.67
PEACE-1, ARASENS	**Triplet**: + docetaxel + abiraterone or darolutamide	OS benefit (ARASENS HR 0.68)

- Docetaxel is also an option with ADT.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

Diagrams to draw

- Testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action.
- Hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

> **Key points to remember:**
> - MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
> - PCTCG 2000: only 2–3% 5-year survival gain.
> - Main use now: flare cover for 21–28 days.
> - ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
> - Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
> - Monotherapy raises testosterone; gynaecomastia two-thirds.
> - EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
> - Withdrawal response 15–30%.

SAQ 4 Saturation biopsy of prostate

P2 · P4 **merged from 2**

Asked as: P2 SAQ "Saturation biopsy in CA prostate" · P4 SAQ "Saturation biopsies of prostate"

Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally (anterior-only cancer 6–20%).

Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation.
- **Focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

Templates and technique

Template	Division	Cores
Original Barzell	Octants (anterior/posterior × apex/base, each side)	4–8 per octant
Modified Barzell	Apex and base of each side × 5 sectors	20
Ginsburg	Anterior, mid, posterior sectors per side	4 per sector

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

Results

- **Initial biopsy:** 12 to 18–21 cores did not raise detection.
- **Repeat setting:** early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route.
- Still used for MRI-negative high suspicion and focal therapy planning (mapping).

Diagram to draw

- Transperineal grid over the prostate with sampling sectors.

> **Key points to remember:**
>
> - Saturation = >12 cores (18–21 or more); 5-mm template mapping.
> - No gain at initial biopsy; used after negatives and for focal therapy.
> - PROMIS: template mapping as reference; mpMRI sensitivity 93%.

SAQ 5 Cryotherapy for CA prostate

P2 · P4 **merged from 2**

Asked as: P4 SAQ "Cryotherapy for carcinoma prostate" · P2 SAQ "Biological effects of cryosurgery"

Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.
- Prostate cryotherapy is **transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

Biological effects: mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

Mechanism	What happens
1. Direct cell injury	*Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals
2. Vascular injury	Endothelial damage, platelet activation and **thrombosis** cause **vascular stasis** and ischaemic necrosis; the main delayed killing effect
3. Apoptosis	Sublethal cold at the periphery triggers programmed cell death
4. Immunomodulation	Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour

- **End result:** coagulative necrosis, then fibrous scar.

Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C and sublethal; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

Indications in prostate cancer

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.

2. **Primary whole-gland:** men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland:** MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; real-time imaging shows the ice ball; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT, randomised controlled trial): **non-inferior** to external beam radiotherapy; worse sexual function.

Complications

	Primary / focal	**Salvage**
ED (erectile dysfunction)	**Almost always** (whole gland)	Very common
Incontinence	Rare	Single digit
Rectourethral fistula	≤0.8% (focal)	1–3.4%
Retention	Commonest after focal	~4%

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD [Cryo On-Line Data] registry).

Comparison with other ablative energies

	Cryotherapy	**HIFU**	**IRE (irreversible electroporation)**
Energy	Freezing	Focused ultrasound heat (≥55 °C)	Electric pulses, non-thermal
Route	Transperineal	Transrectal	Transperineal
Best for	Anterior lesions	Posterior lesions	Lesions near nerves, vessels

Guidelines and advances

- **EAU (European Association of Urology):** primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

Diagram to draw

- Prostate (or tumour) with probe and ice ball: −40 °C isotherm inside the 0 °C edge, and tumour margin.

> **Key points to remember:**
> - Argon freezes, helium thaws (Joule–Thomson); ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm, urethral warmer.
> - Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
> - Main role is salvage: PSA <10, doubling time ≥16 months.
> - Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

SAQ 6 GnRH (LHRH) analogues: agonists and antagonists

P2 · P4 **merged from 2**

Asked as: P4 SAQ "GnRH (LHRH) analogues: agonists and antagonists" (GNRH Analogue; LHRH agonist) · P2 SAQ "LHRH analogue"

Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.

Physiology

- **LHRH (GnRH)** is a hypothalamic decapeptide released in **pulses**. Pulsatile GnRH drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → testosterone (about 95% testicular; about 95% of total).
- **Continuous** exposure **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

GnRH (LHRH) agonists: drugs

Drug	Formulation
Leuprolide	Depot 1, 3, 6 monthly
Goserelin	SC (subcutaneous) implant 1 or 3 monthly
Triptorelin	IM (intramuscular) depot
Histrelin	Implant, 1 year
Buserelin	Nasal or SC

- Substitution at **position 6** of the decapeptide raises potency and half-life; depots last 1 month to 1 year.

Agonists: mechanism

`Agonist → initial LH surge (flare, testosterone up) → receptor down-regulation (1–2 weeks) → LH falls → castrate testosterone in 2–4 weeks`

- Survival **equals orchidectomy**.

Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention.
- Dangerous with spinal or weight-bearing metastases.
- **Prevent with an antiandrogen** (e.g. bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.

- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

Agonist vs antagonist

Agonist	Antagonist
Flare; needs antiandrogen	No flare; castrate by day 3 (degarelix)
FSH rises	FSH falls
Depot to yearly	Monthly SC or oral (relugolix; HERO)

Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better (aim **<20 ng/dL**). Check testosterone when PSA (prostate-specific antigen) rises.

Indications in urology

- Metastatic hormone-sensitive prostate cancer (with ARPI [androgen receptor pathway inhibitor] or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- Recurrent prostate cancer: BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable (PR.7); BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty; cryptorchidism (GnRH or hCG [human chorionic gonadotrophin]: low success, not first-line); endometriosis.

Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).
- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

Diagram to draw

- Hypothalamic–pituitary–testis axis showing agonist and antagonist sites, and the flare.

> **Key points to remember:**
> - Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
> - Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
> - Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
> - Castrate <50 ng/dL; aim <20 ng/dL.

SAQ 7 Robotic radical prostatectomy

P1

Type: SAQ (10 marks)
Source: 2023-Winter Paper 1, Q3b

Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

Diagram to draw: port layout and fascial planes (intra-, inter-, extrafascial).

> **Key points to remember:**
>
> - Surgeon, not platform, decides outcome (Yaxley 2016).
> - Interfascial plane is the standard nerve-sparing plane.
> - Bladder neck stricture <2%; transfusion 1–2%.

SAQ 8 Docetaxel

P1

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 1, Q3b

Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

Uses in urology

Setting	Evidence
mCRPC	**TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive
mHSPC	**CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant
Triplet	ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS
High-risk localised	RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69
Bladder	Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer)
Penis	**TPF** (docetaxel, cisplatin, 5-fluorouracil)

CHAARTED high volume: ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor pathway inhibitor]).
- **Also know:** failed combinations (bevacizumab, atrasentan).

Diagram to draw: microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

> **Key points to remember:**
> - 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
> - CHAARTED: benefit in high-volume mHSPC (HR 0.61).
> - Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

SAQ 9 PIN

P2

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 2, Q3f

Assumed meaning: prostatic intraepithelial neoplasia.

Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

Related lesions

Lesion	Meaning	Action on isolated finding
HGPIN	Weak risk marker	**No repeat biopsy**
PINATYP (HGPIN with adjacent atypical glands)	As "atypical"	Manage as ASAP (atypical small acinar proliferation)
ASAP	Cancer on re-biopsy about 40%	Repeat biopsy ± MRI (magnetic resonance imaging)
AIP	May herald IDC-P	Consider re-biopsy
IDC-P	Usually high-grade cancer present	Definitive treatment; excludes active surveillance

Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

Diagram to draw: benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

> **Key points to remember:**
> - Atypical cells, benign architecture, basal layer retained.
> - Isolated HGPIN needs no repeat biopsy; ASAP does.
> - Cribriform "PIN" is now IDC-P or AIP.

SAQ 10 Gleason's score and its significance

P2

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 2, Q3f

Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

Patterns (modern)

Pattern	Architecture
1–2	Not used on biopsy now
3	Discrete, well-formed glands of any size
4	**Cribriform** (including glomeruloid), poorly formed or fused glands

Pattern	Architecture
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

Grade group	Gleason score	5-year BCR (biochemical recurrence)-free after prostatectomy
1	3+3=6	96%
2	3+4=7	88%
3	4+3=7	63%
4	8 (4+4, 3+5, 5+3)	48%
5	9–10	26%

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

Diagram to draw: Gleason's pattern diagram 3 → 4 → 5.

> **Key points to remember:**
> - Biopsy score = commonest + highest pattern.
> - 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
> - GG4 = score 8; GG5 = 9–10.

SAQ 11 HIFU

P2

Type: SAQ (10 marks)
Source: Practice Set 5 Paper 2, Q3d

Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

Indications

Setting	Use
Focal therapy, prostate cancer	MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis
Whole-gland ablation	Localised cancer; retreatment 29–38%
Salvage after EBRT (external beam radiotherapy)	Biopsy-proven local recurrence, **≥2 years after EBRT**
BPH (benign prostatic hyperplasia)	Approved
Kidney	Investigational (incomplete ablation)

Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

Also know: histotripsy (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

Diagram to draw: transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

> **Key points to remember:**
> - Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
> - Transrectal HIFU suits posterior lesions; TULSA for anterior.
> - Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

SAQ 12 Imaging measures in CA prostate

P2

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 2, Q3e (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.
- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

Sequence	Shows	Cancer
T2-weighted	Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion)	Low signal
DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient)	Cellularity	Bright on DWI, dark on ADC
DCE (dynamic contrast-enhanced)	Vascularity	Early enhancement, washout

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **^{99m}Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

5. Advanced disease

- PSMA PET selects men for **^{177}Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

Diagram to draw: prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
> - High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
> - PSMA nodal sensitivity 30–40%.

SAQ 13 Expectant treatment of CA prostate

P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3d

Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

Two forms of expectant treatment

	AS (active surveillance)	**WW (watchful waiting)**
Patient	Fit for radical therapy, life expectancy ≥10 years	Limited life expectancy (age, comorbidity)
Intent	**Curative, deferred**	**Palliative**
Monitoring	PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy	Infrequent PSA and review; no biopsy
Action	Radical treatment on reclassification	ADT (androgen deprivation therapy) or palliation for symptoms

Evidence

Trial	**Result**
SPCG-4 (pre-PSA)	Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade
PIVOT	No significant mortality benefit; low-risk cancer death 2.7%

Trial	Result
ProtecT (15 years)	Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%)

Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.
- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

Diagram to draw: AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

> **Key points to remember:**
> - AS is deferred cure; WW is palliation.
> - Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
> - ProtecT: no mortality difference at 15 years; more metastases on monitoring.
> - About 25% leave AS by 5 years.

SAQ 14 Molecular staging of CA prostate

P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3e (related: prognosis in 2019-Summer Paper 4, Q3e)

Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.
- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

2. Tissue genomic classifiers

Test	Basis	Predicts
Decipher	22-RNA classifier	Metastasis, cancer death; post-prostatectomy radiotherapy decision
Oncotype DX GPS (Genomic Prostate Score)	17 genes (12 cancer, 5 reference) by RT-PCR	Adverse pathology; AS (active surveillance) choice
Prolaris	31 cell-cycle progression genes	Progression, cancer death

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

Diagram to draw: clinical stage vs molecular stage (tissue, blood, nodes, imaging).

> **Key points to remember:**
> - Molecular staging finds occult disease and predicts behaviour beyond TNM.
> - Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
> - AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
> - Limitation: heterogeneity and mostly retrospective data.

SAQ 15 Prostate cancer prevention trials

P4

Type: SAQ (10 marks)
Source: 2017-Summer Paper 4, Q3a

Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

Landmark trials

	PCPT (2003)	REDUCE (2010)	SELECT (2009)
Agent	Finasteride 5 mg	Dutasteride 0.5 mg	Selenium 200 µg ± vitamin E 400 IU
Men	18,882; PSA (prostate-specific antigen) ≤3	PSA 2.5–10, prior negative biopsy	35,533
Design	7 years; end-of-study biopsy	Biopsy at 2 and 4 years	Stopped early for futility
All cancers	↓24.8%	↓22.8%	No benefit
High grade	Gleason 7–10 ↑ (6.4% vs 5.1%)	Gleason 8–10 ↑ (0.9% vs 0.6%)	Vitamin E alone ↑ risk

The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

Diagram to draw: bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

> **Key points to remember:**
> - PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
> - FDA warning; no approved preventive agent.
> - SELECT negative; vitamin E alone harmful.

SAQ 16 PSMA PET Ct scan

P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3a

Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** ^{68}Ga-PSMA-11 (half-life 68 min); ^{18}F-DCFPyL, ^{18}F-rhPSMA-7.3, ^{18}F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

Indications

1. Primary staging (high risk):

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

2. BCR (biochemical recurrence):

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

3. Intraprostatic detection: PRIMARY trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

4. Theranostics: selects patients for **^{177}Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

Pitfalls

False positive	False negative
Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia	Small nodes (<5 mm)
Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma))	PSMA-negative tumours (about 5–10%)
^{18}F-PSMA-1007 benign bone uptake	Neuroendocrine transformation; masking by bladder activity

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

Diagram to draw: PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

> **Key points to remember:**
> - proPSMA: accuracy 92% vs 65%.
> - BCR: about half positive below PSA 0.5 ng/mL.
> - Nodal sensitivity 30–40%: negative scan does not exclude nodes.
> - VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

SAQ 17 Methods to predict prognosis of patient of Cancer prostate

P4

Type: SAQ (10 marks)
Source: 2019-Summer Paper 4, Q3e

1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

Grade group	Gleason	5-year BCR (biochemical recurrence)-free
1	≤6	96%
2	3+4	88%
3	4+3	63%
4	8	48%
5	9–10	26%

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

2. Risk groups

D'Amico	Low	Intermediate	High
PSA (ng/mL)	≤10	10–20	>20

D'Amico	Low	Intermediate	High
Gleason	≤6	7	8–10
Stage	T1–T2a	T2b	T2c

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

5. Tissue biomarkers

Test	Basis	Use
Decipher	22-RNA classifier	Metastasis risk; post-op radiotherapy
Oncotype DX GPS	17 genes	Adverse pathology; active surveillance
Prolaris	Cell-cycle progression genes	Progression, cancer death

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

Diagram to draw: flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

SAQ 18 PARP inhibitors in Prostate cancer.

P4

Type: SAQ (10 marks)
Source: 2024-Winter Paper 4, Q3f

Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

Key trials

Trial	Setting	Result
PROfound	Olaparib vs enzalutamide/abiraterone after ARPI	BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1)
TRITON3	Rucaparib vs physician's choice	BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit
PROpel	Olaparib + abiraterone, first-line mCRPC	OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA
MAGNITUDE	Niraparib + abiraterone, HRR-altered	BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration
TALAPRO-2	Talazoparib + enzalutamide, first line	All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

Diagram to draw: synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

SAQ 19 Indications of genomic testing in carcinoma prostate

P4

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 4, Q3e (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

Types of test

Type	Sample	What it finds	Use
Germline	Blood or saliva	Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2)	Therapy, family testing, screening
Somatic	Tumour tissue (usually primary) or ctDNA (circulating tumour DNA)	HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency)	Targeted therapy
Gene-expression classifiers	Biopsy or prostatectomy tissue	Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes)	Prognosis

Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).
- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
- **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).

- **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).
- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

Diagram to draw: flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

> **Key points to remember:**
> - Germline: all metastatic, high-risk, family history, Ashkenazi.
> - Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
> - BRCA2: aggressive; avoid casual active surveillance.
> - Decipher, Oncotype GPS, Prolaris: prognosis only.

Rapid Revision Sheet · Section 17

Staging and management of locally advanced (non-organ-confined) CA prostate (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months; P1: 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
- RT + ADT: EORTC 22863 overall survival 79% vs 62%.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival gain.
- After RP: early salvage RT preferred over routine adjuvant RT for most men.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

Metastatic prostate cancer, including newly diagnosed and bone metastases (LAQ)

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: cover the flare with antiandrogen, or use an antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief as longer courses for focal pain; dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance.
- No survival benefit from bone agents.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

Radiotherapy and brachytherapy for CA prostate (LAQ)

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR ^{125}I 145 Gy; HDR ^{192}Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

Castration-resistant prostate cancer (CRPC): definition and treatment (LAQ)

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, ^{177}Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

PSA screening for prostate cancer: evidence, problems and solutions (LAQ)

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

Discuss management of organ confined CA prostate (LAQ)

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

Discuss advances in management of CA prostate (LAQ)

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

TRUS and prostate biopsy (SAQ)

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

Multiparametric MRI, PI-RADS and fusion biopsy (SAQ)

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

Maximum / combined androgen blockade (SAQ)

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

Saturation biopsy of prostate (SAQ)

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

Cryotherapy for CA prostate (SAQ)

- Argon freezes, helium thaws (Joule–Thomson); −40 °C, two cycles, urethral warmer.
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

GnRH (LHRH) analogues: agonists and antagonists (SAQ)

- Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
- Castrate <50 ng/dL; aim <20 ng/dL.

Robotic radical prostatectomy (SAQ)

- Surgeon, not platform, decides outcome (Yaxley 2016).

- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

Docetaxel (SAQ)

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).
- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

PIN (SAQ)

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Gleason's score and its significance (SAQ)

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

HIFU (SAQ)

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Imaging measures in CA prostate (SAQ)

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

Expectant treatment of CA prostate (SAQ)

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

Molecular staging of CA prostate (SAQ)

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Prostate cancer prevention trials (SAQ)

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

PSMA PET Ct scan (SAQ)

- proPSMA: accuracy 92% vs 65%.

- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: ^{177}Lu-PSMA-617 improves OS in mCRPC.

Methods to predict prognosis of patient of Cancer prostate (SAQ)

- Grade group is the strongest predictor.
- D'Amico, NCCN and CAPRA combine PSA, grade and stage.
- Decipher, Oncotype, Prolaris add genomic risk.

PARP inhibitors in Prostate cancer. (SAQ)

- Synthetic lethality; BRCA2 gains most, ATM least.
- PROfound: olaparib, PFS 7.4 vs 3.6 months.
- Combinations: PROpel, MAGNITUDE, TALAPRO-2.

Indications of genomic testing in carcinoma prostate (SAQ)

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.